

Digital Twins in Transportation Infrastructure: An Investigation of the Key Enabling Technologies, Applications, and Challenges

Xiangyu Chang^{ID}, Rui Zhang, Jianxiao Mao, and Yuguang Fu^{ID}

Abstract—Transportation infrastructure constitutes a significant part of civil infrastructure, such as bridges, tunnels, and roads, enormously promoting economic development. Transportation infrastructure is subjected to a high volume of traffic, damage, and component deterioration, as well as disaster events during the service period. Monitoring and management of the transportation infrastructure is always a critical task. Recently, digital twin (DT) is an emerging topic for transportation infrastructure management, with the advancement of artificial intelligence, Internet of Things, big data, and other smart technologies. Herein, DT aims to build a virtual counterpart of the physical infrastructure that is continually updated with its performance, maintenance, and health status. However, research in DT for transportation infrastructure mainly focused on infrastructure management (e.g., traffic state prediction and passenger flow assessment) and ignored the health status of the infrastructure itself. Challenges and associated opportunities exist in the adoption of DT for transportation infrastructure, such as balancing model fidelity and computation efficiency. Thus, it is necessary to investigate the key enabling technologies of DT in transportation infrastructure and provide a comprehensive reference for ongoing and future research. In this paper, a systematic investigation to identify the development of DTs for transportation infrastructure is presented. The paper starts by explaining the definition of DT and highlighting the various characteristics of DT. Next, the key enabling technologies and applications of DT for transportation infrastructure are discussed. Finally, based on the current development status of DT, challenges and open research are discussed along with their potential solutions.

Index Terms—Digital twin, data model integration, operation and maintenance, structural health monitoring, transportation infrastructure.

I. INTRODUCTION

TRANSPORTATION infrastructure has been considered an essential component in offering convenient travel options and encouraging economic growth. The lifecycle of transportation infrastructure includes the design, construction, operation,

maintenance, and disposal stages, and each stage could be divided into several information layers that require effective information exchange mechanisms to allow for interoperability across various levels [1]. In the lifecycle of transportation infrastructures, system integration management during the service period is one of the most critical concerns. To balance actual demands and financial constraints, managers must take into account a variety of intricate structural features, material deterioration, monitoring data, as well as historical and social circumstances [2]. In fact, managers hardly make proper quantitative and data-driven decisions with large amounts of data. They then resort to qualitative analysis to make judgments on the next action, which exposes users to systematic risk and could cause unexpected safety problems [3]. Therefore, substantial effort must be made to ensure efficient information updates and increase decision accuracy during the service period.

Digital twin (DT), associated with advanced techniques, has lately achieved advancements in civil infrastructure. The concept of DT is originally from the aerospace field, and expanded to manufacturing, healthcare, and smart city [4], [5]. Basically, a DT in the scope of civil infrastructure is a cyber counterpart of a physical system, which is continually updated with the performance of the physical entity based on real-time data and information collected, with the goal of providing the health status of the physical system and proactive maintenance strategies [6]. Originally, the digital shadow was used before the concept of DT e.g., real-time simulation and assessment for decision-making processes [7]. The digital shadow is to be understood as a pure assignment of the status data by a measurement to a specific asset or system at a specific instant. It has no claim to any knowledge generation from a scientific perspective, but forms the basis for a statistical analysis between different instances of an asset or physical system [8], [9]. In contrast, the DT can accurately reflect and update the status of the physical system as sensors collect and transmit real-time data to the virtual system due to the advancement in computing and internet of things (IoT) technology [10], [11].

DT technology has been widely applied in various fields for its high potential to accurately represent and manage physical systems [12], [13], [14]. The main fields include manufacturing, building management, and health care. The first recognized usage of DT is in the manufacturing industry. The manufacturing industry has used DT for automating prod-

Manuscript received 14 February 2023; revised 9 October 2023 and 13 March 2024; accepted 2 May 2024. Date of publication 27 May 2024; date of current version 2 July 2024. This work was supported in part by the National Research Foundation, Singapore, under its AI Singapore Program (AISG) under Award AISG2-TC-2021-001; and in part by the Ministry of Education Tier 1 Grants, Singapore, under Grant RS04/21 and Grant RG121/21. The Associate Editor for this article was B. F. Ciuffo. (Corresponding author: Yuguang Fu.)

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Digital Object Identifier 10.1109/TITS.2024.3401716

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uct manufacture, driving shorter procedures, data analytics, and product quality improvement [15], [16]. For example, Knapp et al. [17] developed building blocks for modeling the DT of additive manufacturing to evaluate the impact of process factors on cooling rates and other structural properties. Furthermore, DT was applied in curtailing the production of unwanted items for sustainable construction [18]. Applications of DT are becoming popular in building management. DT is crucial to many aspects of built-environment projects, such as asset creation, delivery, and management [19]. The data collected from sensors can help managers evaluate the structural health status of buildings and make decisions [20]. For example, Francisco et al. [21] developed a DT system of building performance evaluation by defining temporally segmented, daily building energy benchmarks and measuring their quantitative variations, and investigated the influence of different strategies on near real-time energy management. Another area where DT is widely used is health care. DT technologies were employed in several studies to plan and conduct surgical operations [22], [23], as well as assess and forecast a patient's health [24]. Furthermore, DT may be utilized in combination with AI algorithms to make better forecasts and judgments [25], [26]. Although applications of DT in these fields are still in nascent stages, the potential of DT is vast in the future [5], [27].

Likewise, the DT has significant potential to resolve critical challenges in the management of transportation infrastructure. The paper presents an overview of recent developments in DT technology and concepts, and investigates their potential applications and challenges for transportation infrastructure. A total of 143 papers are analyzed in order to provide a comprehensive review of previous research on the developments and applications of DT in transportation infrastructure. The rest of this paper is organized as follows: Section II will describe DT and its framework, and identify related concepts and applications while highlighting the evolution stages of DT. Section III summarizes the current state-of-art technologies and approaches for developing virtual systems. Section IV describes challenges and expected features for DT research in transportation infrastructure. Section V presents the conclusions.

II. DIGITAL TWIN OVERVIEW

A. Digital Twin Definition

The concept of DT was initially introduced by Professor Grieves of the University of Michigan in 2003 and originated from product life cycle management. Then, the US Department of Defense applied the idea of DT to concerns such as spacecraft health maintenance in product life management [28]. The National Aeronautics and Space Administration (NASA) defines DT as a multiphysics, multiscale, probabilistic simulation that employs physical models, sensor updates, fleet history, etc. in order to simulate the life of its twin [29]. Although DT is gaining popularity in academic literature and industrial practice, there is no commonly accepted definition for it. Several explanations and definitions of DT have been

proposed in various fields. Table I provides a summary of different definitions of DT.

According to these explanations and definitions of DT, although the specific definitions vary, the general characteristics are similar. The following characteristics of DT are summarized:

1) Real-time reflection. In DT, two systems exist, including a physical system and a virtual counterpart. In the form of digitization, the DT creates a high-fidelity virtual model that is identical to the physical system [39]. In other words, the virtual system is updated in real-time to represent the physical one.

2) Digital modeling and simulation. Accurate models provide the foundation for making appropriate decisions in DT systems. At the beginning of building a DT system, a high-fidelity model is built to reflect the health status of the physical system. Then, the high-fidelity model can experience continuous improvements with the physical system in parallel based on the data and decisions.

3) Integration and interaction. DT is a full-flow, full-element, and full-service integration system. A DT usually consists of several subsystems, establishing the foundation for high-precision modeling. The data is generated and collected for physical system information from different subsystems at various stages in DT [40].

4) Decision and service. DT has real-time data and all historical information for a physical system on one platform throughout its life cycle. Through advanced data analysis tools, the DT has significant potential to make more informed and effective decisions and services.

In summary, a DT in the scope of civil infrastructure is a cyber counterpart of a physical system, which is continually updated with the performance of the physical entity based on real-time data and information collected, with the goal of providing the health status of the physical system and proactive maintenance strategies.

B. Publication Screening and Scientometric Analysis

This paper aims at investigating the DT application in transportation infrastructure, which currently falls behind the manufacturing and construction sectors in terms of DT maturity. The process for paper searching is summarized:

- 1) Databases and platforms used: The Web of Science, Scopus, and Google Scholar tools as the three most used databases were used as the search engines.
- 2) Search strategy: The search included terms relevant to the broad research domains, such as road digital twin, bridge digital twin, tunnel digital twin, aviation digital twin, transportation infrastructure, port, airport, etc. Search strings combined keywords using AND and OR, e.g., "Digital Twin" AND ("Transportation" OR "Transport" OR "Bridge" OR "Tunnel" OR "Road" OR "Highway" OR "Railway" OR "Airport" OR "Port"). To avoid the ambiguity of the word 'bridge', additional context-specific keywords and Boolean logic that narrows down the focus to DTs specifically applied to bridge infrastructure is added, such as "Digital Twin"

TABLE I
COMPARISON OF DIFFERENT DEFINITIONS OF DT

Reference	Definition	Area
Knapp <i>et al.</i> [17]	An integrated multiphysics, multiscale, probabilistic simulation of a vehicle or system that uses the best available physical models, sensor updates, and fleet history, among others, to mirror the life of its flying twin. The digital twin is ultrarealistic and may consider one or more important and interdependent vehicle systems.	Manufacturing
Chen [30]	A digital twin is a computerized model of a physical device or system that represents all functional features and links with the working elements.	Manufacturing
Madni <i>et al.</i> [31]	A digital twin is a virtual instance of a physical system (twin) that is continually updated with the latter's performance, maintenance, and health status data throughout the physical system's life cycle.	Manufacturing
Zheng <i>et al.</i> [32]	DT is an integrated system that can simulate, monitor, calculate, regulate, and control the system status and process. It has the characteristics of individualization, high efficiency, and highly quasi-real.	Product life cycle management
Vrabic <i>et al.</i> [33]	A digital twin is a digital representation of a physical item or assembly using integrated simulations and service data. The digital representation holds information from multiple sources across the product lifecycle.	Product life cycle management
Ramu <i>et al.</i> [34]	A digital twin is a combination of a computational model and a real-world system. It allows customers and engineers to monitor, visualize, analyze, and optimize operational assets, processes, and systems by manipulating live IoT data besides providing real-time insights into performance and activity	Smart cities
Dang <i>et al.</i> [2]	A digital twin is a framework aiming to perform continuous monitoring and proactive maintenance through continuous data from physical entities to virtual counterparts. monitoring service evolves from periodical, generic, and physics-based models to real-time, personalized, and data-driven ones, thus optimizing maintenance strategy, increasing reliability, and safety of the structure, and extending its remaining service life.	Structural health monitoring
Yu <i>et al.</i> [35]	Digital twin is a mapping relationship between the physical space and virtual space and is intelligent. It maps, records, simulates, predicts, and manages the running track of the life cycle of objects in the physical world and digital virtual space.	Structural health monitoring
Kim and Kim [36]	A digital twin is a digital version of a physical twin and refers to the total sum of information generated from the interworking of digital physics rather than being a simple digital representation.	Infrastructure management
Han <i>et al.</i> [37]	A Digital Twin is a high-fidelity representation of the operational dynamics of its physical counterpart, enabled by near real-time synchronization between the cyberspace and physical space.	Infrastructure management
Reifsnider and Majumdar [38]	A digital twin is a kind of ultra-high fidelity simulation integrating with an on-board health management system, maintenance history, and historical vehicle and fleet data. It can mirror the whole life of a specific flying physical twin (or tail number), which enables significant gains in safety and reliability.	Composite materials
Liu <i>et al.</i> [6]	The digital twin is actually a living model of the physical asset or system, which continually adapts to operational changes based on the collected online data and information, and can forecast the future of the corresponding physical counterpart	Maintenance

AND ("Bridge" AND ("Maintenance" OR "Monitoring" OR "Measurement" OR "Measuring" OR "Assessment" OR "Prediction" OR "Management". This phase entails a detailed examination of the title and abstract of each article. The search results were sorted by relevance.

- 3) Inclusion and exclusion criteria: Articles were included if they: a) were published between 2010 and 2023; b) were peer-reviewed articles and conference papers. Articles were excluded if they were CORRECTION, PATENT, NOTE, BOOK CHAPTER, LETTER.

- 4) Screening process: Initially, search results were screened by title and abstract using predefined keywords and criteria. Following this automated filtering, we conducted a thorough manual review of full text for each article that passed the initial screening. This manual process involved a detailed assessment against our inclusion and exclusion criteria, ensuring the relevance and quality of the studies included in our review.

The initial search yielded 712 papers. After the application of exclusion and inclusion criteria, 143 papers were retained

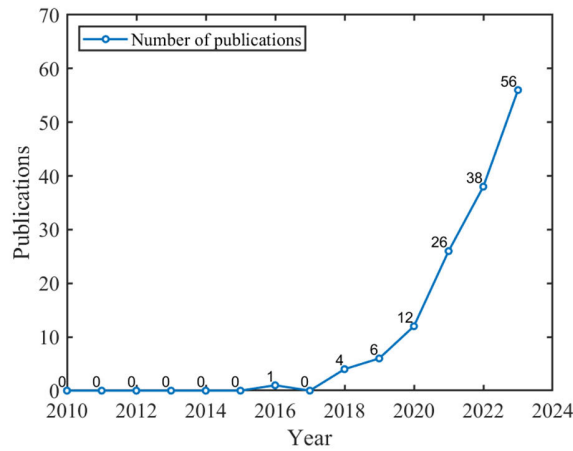


Fig. 1. Developing trend of DT in transportation infrastructure based on selected publications.

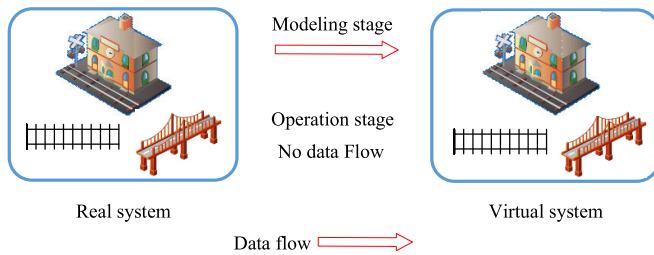


Fig. 2. Data flow of a digital model.

for the final review. Figure 1 presents the yearly distribution of the 143 papers examined in this study, offering a comprehensive perspective on the evolution of DT applications in transportation infrastructure starting from 2010. The trend line shows that the adoption of DT applications is fairly consistent and limited up to 2017. After this period, there is a noticeable yearly growth in scholarly publications, signifying an increasing focus and interest in the DT concept among researchers.

During the screening process, this section also discussed several related concepts of DTs, including digital model, digital shadow, and DT. After clarification and comparison, we provide a summary of all usages of DT concepts in transportation infrastructure. From our perspective, the digital model and digital shadow are all early evolvement stages of DT. Therefore, in the following review sections, we also include some related papers.

1) *Digital Model*: A digital model is a digital representation of a current or planned physical system. More crucially, there is no data interaction between the digital model and the physical system after it is built [5]. This means that changes to the physical system have no influence on the digital model after the model is developed. Figure 2 illustrates the data flow in a digital model.

2) *Digital Shadow*: A digital shadow is a digital version of a physical system designed to generate a real-time assessment based on all available data [41]. An important feature of a digital shadow is that it only allows one-way interaction between physical and digital systems. Changes in the physical

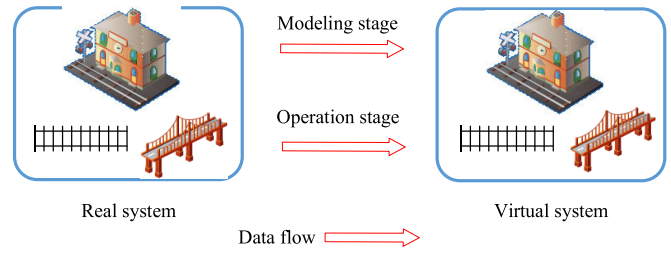


Fig. 3. Data flow of a digital shadow.

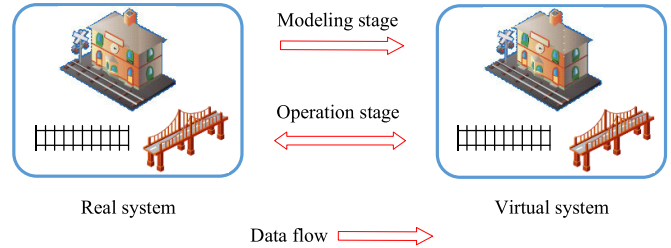


Fig. 4. Data flow of a digital twin.

system could result in changes in the virtual system, but not vice versa. Figure 3 illustrates the data flow in a digital shadow.

In contrast, DT fully integrates data flow between physical and digital objects in both directions throughout the lifecycle. Changes that happen in these two systems automatically lead to changes toward each other. Figure 4 illustrates the data flow in a DT.

After publication screening of the 143 papers, the scientometric analysis is conducted to provide information, including main research interests, the co-occurrence of keywords, citation ranks of publication sources, and co-authorship between countries. Science mapping is conducted herein, which is utilized to analyze and visualize the structure of a research area and the connections between items [42]. The science mapping tool used in this study is VOSviewer (v1.6.19). Furthermore, the main research interests derived from the scientometric analysis can help authors summarize the key enabling technologies for DT and their applications in transportation infrastructure.

In general, the keywords indicate the authors' research focus and key interests in each paper. To search for the main research interests in DT in transportation infrastructure, a co-occurrence analysis of all keywords from the screened publications is investigated. The VOSviewer generates a co-occurrence network (Figure 5) of all relevant keywords. And the network is generated by selecting 'All keywords' and 'Full counting' as the types of 'Unit of analysis' and 'Counting method', respectively. Besides, the 'Minimum number of occurrences of a keyword' is set to '4'. The items with high-strength links are clustered with a non-overlapping curve in different colors. Only the top 54 keywords are chosen and appeared in the network. The size of an item stands for the popularity of occurrence [43]. Figure 5 shows the co-occurrence network with different cluster colors. Five clusters are generated from the selected keywords, and the details for the clusters and related keywords are summarized in Table II. And the technology-related keywords are selected to identify

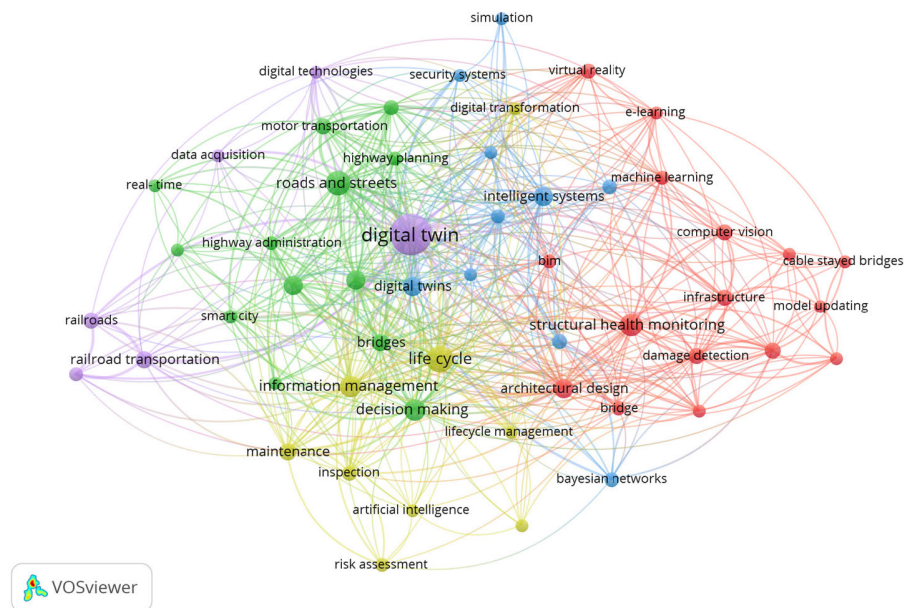


Fig. 5. Co-occurrence network of keywords.

the crucial technologies that can make DT progress and develop. Based on the rank of ‘Occurrences’, the most popular keywords can be selected as having the highest number of the ‘Occurrences’. It can be concluded that ‘Decision making’, ‘Finite element method’, ‘Computer vision’, ‘Bayesian networks’, and ‘Deep learning’ are among the most popular keywords in VOSviewer analysis. They are the key enabling technologies to make the DT concept a reality. A more comprehensive analysis for the key enabling technologies will be conducted in Section III, relying on the different DT layers. From the co-occurrence network, it can also be concluded that DT is becoming popular in transportation infrastructure, such as the bridge system.

Citation analysis of the screened papers can help readers gain a deep insight into the sources of related information in this field of study. Journals or conference proceedings with more documents and citations have greater influence in related research areas. The top 5 journals are summarized in Table III. Due to the limitations of the data resources, the citation numbers are not large and cannot reflect the popularity of the related research fields. The main reason is that ‘Digital Twins in Transportation Infrastructure’ is a relatively new topic with a moderate number of publications available for citation analysis.

The active countries that participated in the selected research areas can be identified by co-authorship analysis of countries and the established collaboration network. By setting the minimum number of documents (i.e., publications) of a country to 2, the collaboration network of countries is then shown in Figure 6. The top five most active countries in the selected research area are China, the United States, the United Kingdom, Italy, and Sweden.

C. Digital Twin Framework

DTs establish a dynamic virtual model of a physical entity in a digital manner, simulating and depicting the

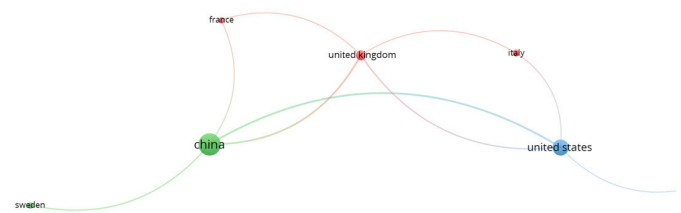


Fig. 6. Collaboration network of countries.

behaviors, and rules of the physical entity in the real environment across multiple dimensions, multiple spatiotemporal scales, multiple disciplines, and multiple physical quantities [44]. In particular, DTs serve as a comprehensive framework that establishes a connection between physical and virtual systems. Transportation infrastructure constitutes a pivotal element within contemporary societies, encompassing diverse transportation modes such as roadways, railways, bridges, tunnels, and public transit systems. The transportation infrastructure system holds paramount significance in supporting economic development, augmenting connectivity, and elevating the overall quality of life for communities. The extensive integration of the IoT facilitates the acquisition and transmission of data from diverse sources within infrastructure systems, promoting interconnection among physical systems. The virtual system, regarded as a subsystem, seamlessly incorporates diverse digital analyses and services, such as digital modeling layer, data fusion layer, service layer, and interaction with people [12]. Hence, a complete DT architecture for transportation infrastructure usually consists of six layers: data acquisition layer, transmission layer, digital modeling layer, data fusion layer, service layer, and interaction with people, which are discussed in the following part and illustrated in Figure 7.

TABLE II
SUMMARY OF CLUSTERS GENERATED FROM THE SELECTED KEYWORDS

Cluster	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5
Color	Red	Green	Blue	Yellow	Purple
Area	Right side of the map	Middle of the map	Middle of the map	Bottom of the map	Left side of the map
Related keywords	1.Architectural design	1.Bridges			
	2.BIM	2.Decision making			
	3.Bridge	3.Highway	1. Bayesian networks		
	4.Cable stayed Bridges	4.Highway administration	2.Deep learning	1.Artificial intelligence	
	5.Computer vision	5.Internet of things	3.Digital modeling	2.Building information	1.Data acquisition
	6.Damage detection	6.Motor transportation	4.Digital twins	3.Digital transformation	2.Digital technologies
	7.E-learning	7.Performance	5.Forecasting	4.Information management	3.Digital twin
	8.Finite element method	8.Real-time	6.Intelligent systems	5.Inspection	4.Railroad transportation
	9.Infrastructure	9.Road infrastructures	7.Intelligent transportation	6.Life cycle	5.Railroads
	10.Machine learning	10. Roads and streets	8.Intelligent vehicle	7.Lifecycle management	6.Railway infrastructure
	11.Model updating	11.Smart city	9.Security systems	8.Maintenance	
	12.Numerical models	12.Sustainable development	10.Simulation	9.Risk assessment	
	13.Structural health monitoring	13.Transportation infrastructure			
	14.Virtual reality				

1) *Data Acquisition Layer*: The data acquisition layer is tasked with consistently acquiring and collecting real-world data from physical assets, processes, or systems to ensure the DT maintains a precise representation of its real-world counterpart. Within transportation infrastructure systems, an array of sensors and IoT devices are deployed to monitor and gather data on their operational performance, condition, and surrounding environment. These sensors effectively measure an array of parameters, including but not limited to temperature, pressure, vibration, humidity, location, and other pertinent variables. The design of a data collection system is a crucial and challenging problem due to the heterogeneity and enormous volume of data [45]. Common data collection techniques are contactless data collection (e.g., image-based

techniques), distributed sensor systems (e.g., fiber optic sensors), and wireless smart sensors.

2) *Transmission Layer*: The data transmission layer is responsible for facilitating the efficient and secure transfer of data between the physical system and its corresponding virtual representation. It guarantees that data generated from sensors and IoT devices in the real world is seamlessly transferred to the DT platform, allowing for its data processing and decision-making processes. Diverse connectivity options are provided, catering to various device types and environments, including traditional wire transmission, short-range coverage technologies (e.g., WiFi and mobile-to-mobile), wider coverage technologies (e.g., 4G, long-term evolution, and 5G), or even satellite communication for remote assets. In some

TABLE III
SUMMARY OF TOP FIVE RESEARCH SOURCES

Journal	Number of documents	Citations	Total link strength
Sustainability (Switzerland)	5	91	2
IEEE Transactions on Intelligent Transportation Systems	4	10	0
IEEE Access	4	23	1
Infrastructures	3	10	0
Engineering structures	3	339	1

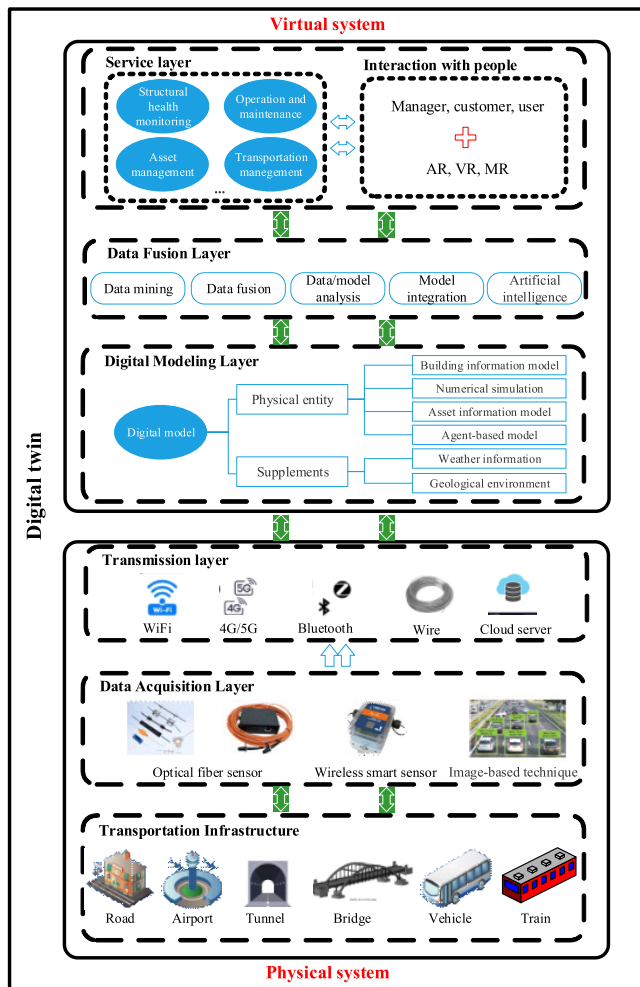


Fig. 7. Illustration of a DT framework for transportation infrastructure.

cases, the integration of edge computing within the data transmission layer allows data processing and analysis in close proximity to the data source (at the edge of the network) before transmitting it to the central DT platform. This approach

effectively reduces latency and minimizes the volume of data sent over the network, which can be particularly beneficial for real-time applications.

3) *Digital Modeling Layer*: The digital modeling layer serves as the critical component responsible for modeling and managing the virtual representation of physical systems. Typically, this process involves creating a detailed and dynamic model that faithfully emulates its real-world counterpart, encompassing its geometry, behavior, characteristics, and interactions within its environment. Examples of digital representations of physical entities are building information model (BIM), physics-based models, and data-driven models [46]. More specifically, three-dimensional (3D) models (e.g., BIM and geographic information system) furnish a visual and spatial comprehension of the structure and its components. Physics-based models can simulate structural behavior and responses to diverse inputs and conditions. Real-time data from sensors and actuators can be used to develop data-driven models, emulating and predicting the behavior of the physical system. The model types of digital representations are determined by application scenarios [47]. Once the type and model details of the DT system are designed, well-organized modeling and verifying processes are essential to ensure the DT is in line with the target physical system [12], [34].

4) *Data Fusion Layer*: The data fusion layer plays a critical role in facilitating the integration of real-world data from the data acquisition layer with the virtual models created in the digital modeling layer. This integration process ensures that the DT precisely mirrors the current state and behavior of the physical systems and can be used effectively for data analysis and decision-making. This layer usually contains data processing, data mapping, data synchronization, data fusion, simulation, domain knowledge from experts, and AI-based learning. Different data and models are obtained by various approaches and represent information from various parts and layers of the physical system. Based on all the available data, different technologies of data and model analysis can be realized for decision-making, such as anomaly detection, data prediction, and data reconstruction. The common methods of integration include rule-based methods, machine learning (ML), DL, and model updating methods [48]. By providing informed decisions, this layer empowers stakeholders to make data-driven choices aimed at enhancing efficiency, minimizing downtime, and optimizing overall operations.

5) *Service Layer*: The service layer refers to the interface or collection of services offered by the DT platform, catering to diverse functionalities, interactions, and applications associated with the virtual counterpart of the physical system. Serving as an intermediary, this layer establishes a linkage between the underlying data, models, and functionalities of the DT and the external applications, users, or systems engaging with it. Common service functions for transportation infrastructure include bridge health monitoring, damage detection, lifecycle management, safety assessment, and proactive maintenance strategy. For instance, Dan et al. [49] presented a DT system for bridges using monitored traffic loads to provide operational condition assessments and safety risks in the transportation network. In addition, human satisfaction and

evaluation of the constructed DT system are directly influenced by social services, which means that feedback from people could be used as external information to improve the DT system [12].

6) *Interaction With People*: DT system serves not just as a technical or informational tool, but as an interactive platform that supports decision-making, operation, and the overall user experience. Interaction with people transforms the way individuals engage with physical assets and systems, allowing for more intuitive and data-driven insights, real-time monitoring, and informed decision-making. As the areas of application expand, DTs need to accommodate the application requirements from different fields and levels of users (such as field maintenance workers, technical experts, management decision-makers, and users) and various business needs [44], [50]. For example, in the design stage of transportation infrastructure, the service layer is mainly targeted at professionals, providing them with suggestions for decision-making [13]. Professionals can monitor real-time data from sensors and IoT devices in the physical system through the DT platform by using visualization tools, such as dashboards and charts. However, effective information visualization and communication tools should be provided during the implementation stage to help understand the available information for ordinary managers and users [51]. The implementation of user-friendly interfaces, such as web applications, mobile apps, or augmented reality (AR) interfaces, can enhance accessibility and intuitive user experiences.

III. KEY ENABLING TECHNOLOGIES

The emergence of several underlying technologies has made DT a reality. This section discusses the key enabling technologies for modeling a fully functional DT system in the field of transportation infrastructure.

A. Digital Modeling Layer

The physical system is generally represented by a set of digital models in the digital modeling layer. In the intelligent building environment, BIM has proven to be a powerful digital modeling technique to integrate building information throughout the life cycle of buildings. However, BIM is mainly used in the building industry and is not fully applicable to transportation infrastructure. This section reviews several critical modelling techniques applicable to transportation infrastructure, including mechanical modelling, GIS, and city information modeling (CIM). Figure 8 shows the applications of digital modeling technologies in DT.

1) *Mechanical Modelling*: Mechanical modelling is fundamental and crucial to the design and management of infrastructure structures. It models a virtual representation of one or several components (e.g., bridges, tunnels, viaducts, and roads) based on structural knowledge and mechanisms. The advantages of this technique include high efficiency, rapid convergence, and low cost [52]. Advanced computer-aided design software facilitates the generation of detailed 3D models characterized by precise geometry, dimensions, and material attributes. Numerical simulations provide the tools to

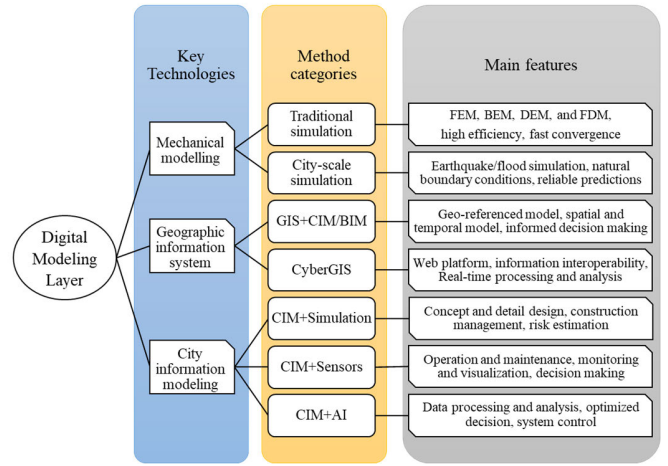


Fig. 8. Technologies of digital modeling layer in DT.

anticipate the behavior of mechanical systems across diverse scenarios. Integration of various simulation tools enables seamless and real-time analysis of stress, heat transfer, and fluid dynamics within the DT. Moreover, the implementation of parametric modeling facilitates the development of models through effortless modification and revision of parameter adjustments. Notably, generative design utilities employ mechanical modeling techniques and AI algorithms to autonomously devise optimized design resolutions guided by predetermined criteria [53]. Typical mechanical modelling techniques include FEM, DEM, BEM, FDM, and computational fluid dynamics (CFD) [54], [55]. Nonetheless, it encounters the challenges of time-consuming modeling, computational cost and time, and continuous verification, when it is scaled up, e.g., to simulate a network of transportation infrastructure.

With significant developments in high-performance computing and parallel computing technologies, numerical simulation at the city scale is now technologically viable, allowing for high-resolution results for a whole city at an affordable computational cost. It solves the traditional problems in the simulation of localized domains, such as unrealistic domain borders, breaking off connections between the interior and exterior of models at boundaries, and inaccurate predictions [56].

2) *Geographic Information System*: GIS techniques are used to analyze 3D geographic information collected from many sources (e.g., maps, satellite imagery, and geospatial data) in order to get a better understanding of its present and future patterns, as well as its impact on physical entities [57]. GIS has the capability to model topography, surrounding environment, land utilization, and additional geographic attributes. Accurate geospatial data can be obtained through high-resolution satellite images and light detection and ranging (LiDAR) data, ensuring detailed and precise geographical insights. Subsequently, GIS platforms provide 3D visualization and integration with IoT data, facilitating the monitoring and management of the physical system [58]. In transportation infrastructure, GIS can bring more advantages to the visualization and analysis of the data sets, such

as a broader and more realistic perspective of 3D geographic environments with vegetation, terrain, and road networks [59].

GIS has been a popular technique for transportation infrastructure networks, and several recent advances have made it more comprehensive and powerful. BIM models can be appropriately connected with the surrounding environment in GIS models, such as strata, groundwater, and roads, in spatial analysis or visualization. Integration of BIM and GIS provides critical quantitative analytical information for data mining and informed decisions, as well as visualization opportunities [60]. In reality, maintaining up-to-date geospatial data and accurately aligning it with the digital model requires continuous updates and synchronization. Hence, another direction is the integration of cyber-physical systems and GIS (CyberGIS), which provides a new opportunity for online spatial analysis. In practice, CyberGIS develops a web platform for GIS instead of only running on a single computer, allowing it to handle the interoperability of geospatial information and processing software [61], [62]. CyberGIS could also provide real-time information about the geospatial data and results of data processing and analysis.

3) *City Information Modeling*: CIM has been regarded as a rich digital technology that uses an object-oriented approach to describe the properties of a structure at the city scale [59]. CIM extends the idea of BIM to broader urban areas, allowing the development of virtual city models that include urban infrastructure, transportation networks, utilities, and other linked urban components. A CIM model, as a 3D digital representation, combines geometric and semantic information on structural elements, providing facility managers with a virtual collaborative platform for retrieving, analyzing, and processing various forms of data with better visibility and efficiency [1]. In the field of transportation infrastructure, CIM has been shown to be a parametric digital modeling technology capable of supporting infrastructures over their entire life cycle [63], [64]. Examples for the tools of CIM include Revit, BIMFACE, and Unity3D software. CIM provides facility managers with a visual platform to display and analyze the collected data. Due to the sharable and consistent database, CIM has been used in various aspects, such as supporting urban planning, infrastructure development, disaster management, and sustainability assessments [65], [66].

Although CIM is a promising solution for DT modelling of transportation infrastructure, it cannot address all challenges. Because CIM alone only provides static data of structures, it is unable to automatically update real-time information in the established model [67]. As the advancement of IoT sensors, cloud servers, and high-performance computers, sensing devices are networked and allow information sharing across different platforms [68]. Then, real-time sensing data processing and visualization become accessible in CIM models, and these models can be updated over time once the latest data from structures become available [66], [69], [70].

4) *Development and Challenges*: The digital modeling layer establishes a detailed virtual portrayal within a DT, mirroring the geometrical, behavioral, and characteristic attributes of the physical structure, as well as its interactions with the surroundings. The progression of enabling technologies, such

Rule-based method	Machine learning	Deep learning	Computer vision	Model updating
• White box • 'If-then' rules • Explicit representation • Transparent	• SVM, GPR • Supervised and unsupervised learning • Damage identification, pattern recognition	• Neural networks • Better prediction accuracy • Data mining	• Image and videos correlation problems • Contactless measurements • Difficult-to-access structures	• Data-driven model, physics-based model • Understand the structural behavior • Informed decisions

Fig. 9. Technologies of data fusion layer in DT.

as mechanical modeling, GIS, and CIM, is propelling the generation of progressively detailed, precise, and interlinked virtual counterparts of physical systems. While advancements are promising, continuous endeavors are imperative to tackle challenges associated with precision, integration, and scalability of models. Addressing these challenges is critical to realizing the full potential of digital modeling technology for precise modeling, simulation, and optimization of physical assets and systems.

B. Data Fusion Layer

Data fusion is an interdisciplinary subject that involves various fields, with a focus on data analysis and its integration into models to acquire a better understanding of the physical system [5]. This section introduces the several tools and technologies needed for data fusion and analysis. Typical examples of technologies include rule-based method, ML, DL, CV, and model updating. These techniques are discussed further below. Figure 9 shows the applications of data fusion technologies in DT.

1) *Rule-Based Method*: Rule-based methods process data based on domain knowledge from experts. It includes a set of rules that consider all interactions in a physical system [71]. Rules are defined to interpret and transform data obtained from sensors and IoT devices on the physical system into the virtual system's corresponding variables or parameters. The most significant advantage of a rule-based system is the explicit representation and interaction of objects. The system is also transparent due to its clear expression, which allows for minimal barriers to the verification of the system [72]. Various rule-based systems have been built to assist decision analysis and have grown to be essential DT tools for transportation infrastructure [73]. For example, Dimitrova et al. [74] utilized semantic web technologies for knowledge capture and reasoning to develop a decision support system that helps with pathology diagnosis and tunnel evaluation. An interdisciplinary group of tunnel specialists and knowledge engineers created the system in real-world situations provided by an actual project. However, compared with AI-based methods, rule-based methods also lack self-learning mechanisms in decision-making processes [75].

2) *Machine Learning*: ML is a part of AI that allows computers to solve complicated problems and process data without human intervention [76]. ML algorithms have the capacity to acquire and deduce the relationship between sensor data and the corresponding parameters within the virtual framework, particularly in scenarios involving multiple sensors and variables. ML can automate various intermediate

processes of data fusion, thereby diminishing the necessity for manual intervention [77]. Based on the requirement of data labeling, ML techniques can be divided into supervised, unsupervised, and semi-supervised learning algorithms [78], [79]. ML algorithms can be used to identify structural damage and degradation (e.g., corrosion, carbonation, fatigue, etc.) for transportation infrastructure, because monitoring data is usually associated with degradation mechanisms [3]. Ren et al. [80] used SVM algorithm to identify damaged cables in a cable-stayed bridge based on deck bending strain data. ML algorithms are also powerful tools to predict structural behavior and provide dynamic feedback on traffic for predictive maintenance of structures. Naraniecki et al. [81] proposed a data-driven approach to predict the condition states of railway bridges using inspection data based on CIM and decision tree algorithm. The prediction results offer a foundation for accurate maintenance management actions.

3) *Deep Learning*: DL is another part of AI. DL techniques use multilayer neural networks with autonomous input feature extraction to learn internal relationships between various types of information [82]. Compared to ML, DL algorithms have complex architectures with more layers (generally more than three) and require more data and a longer time for training. As a result, DL algorithms use complex structures to discover relationships, patterns, and representations from data, providing more advanced integration techniques and improved prediction accuracy. When working with high-dimensional data or complicated sensor inputs, DL algorithms can be especially effective. Another feature of DL is transfer learning. Pre-trained DL models may be fine-tuned for specific integration tasks by utilizing information from relevant fields. DL has been adopted for DT applications. For instance, Dang et al. [2] presented a DT framework for structural health monitoring based on cloud computing and DL algorithms to execute real-time monitoring and provide proactive maintenance strategies efficiently. Utilization of cloud computing infrastructure and a user-friendly web application improves the data interaction between physical structure, digital model, and human interventions. These algorithms are also applied for data mining, data imputation, damage identification, and anomaly detection, like ML algorithms [83]. Compared to ML, the application of DL in DT for transportation infrastructure is still limited.

4) *Computer Vision*: CV is an important technology for visual data processing and provides promising methods for image-related problems. CV algorithms are capable of processing images or videos obtained by cameras or sensors on the physical system and extracting useful information, such as objects, spatial attributes, and environmental conditions [84]. This information can additionally be incorporated into the DT to augment its comprehension of the physical system and offer predictive maintenance strategies. When compared to conventional methods, the CV technique is more convenient, efficient, and cost-effective, particularly for difficult-to-access structures. In transportation infrastructure, the CV technique can be applied for deformation measurement and damage monitoring by image recognition algorithms. Because of their characteristics of contactless measurements, multiple point

measurements, and high precision, these techniques have gained space in the aspect of monitoring infrastructures [85]. However, the utilization of CV techniques must take into account potential obstacles, including fluctuations in lighting conditions, obstructions, and alterations in the physical system occurring over time.

5) *Model Updating*: Accurate models provide the foundation for making appropriate decisions. Model updating based on additional data/information has lately become a popular method to ensure the quality of a digital model. Model updating means applying information extracted from previously observed data to verify and update the constructed model [86]. Model updating methods have two main types: data-driven techniques and finite element (FE) model updating techniques. The data-driven model is updated according to information derived from monitoring data and/or physics-based relationships to solve problems. For the FE model updating technique, the structural features (e.g., natural frequencies and mode shapes) of the initial FE model are updated by minimizing the differences between the observed and simulated data [87]. With sufficient monitoring and experimental data, model updating methods could help engineers understand the information of structures and make informed decisions.

With the development and maturity of IoT and AI technologies, many studies have been performed in order to establish methods for structural response prediction and damage diagnosis employing model updating methods [88], [89]. Yang et al. [90] proposed a physics-guided machine learning-based video processing method that can identify the modal parameters of cable vibration based on the video of cables for cable-stayed bridges. This approach provided a low-cost and noncontact technique to estimate the cable tension using only the video of the vibrating cable. Using a FE model update strategy, Zhang and Lin [54] employed CV algorithms and geometric analyses to develop a FE model updating methodology for updating the geometric changes induced by structural defects. This study automated the pipeline from structural damage identification to numerical modeling for rapid structural capacity assessment.

6) *Development and Challenges*: In summary, the data fusion layer is critical to establishing an accurate mapping between the real system and its virtual counterpart. The increasing utilization of IoT devices and sensors gives rise to extensive volumes of data. Emerging data forms, such as images and videos, coupled with associated technologies, offer knowledgeable observations and insights for the physical system. Progress in edge computing and communication methodologies further enable more efficient and real-time data integration. ML and DL approaches are being used to improve data processing, data imputation, anomaly detection, and model updates. However, issues like data quality, scalability, interpretability, and ensuring correct model updates continue to be important areas of study and development.

C. Service and Interaction With People

In the DT framework, the service layer interprets the information from the data and model analysis to provide

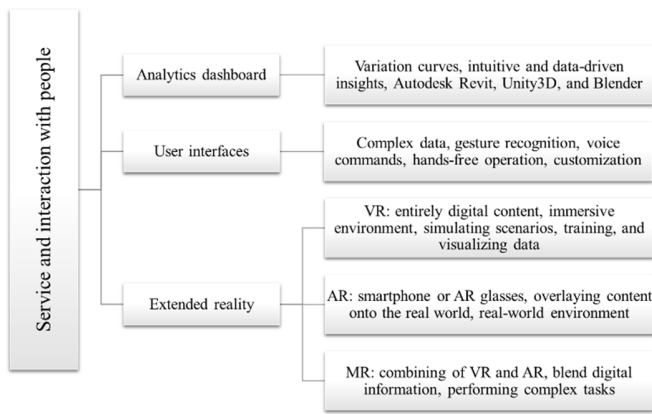


Fig. 10. Technologies of service and interaction with people layers in DT.

social services. Tools for interaction (such as user interfaces, visualization utilities, and communication channels) enable stakeholders to access and engage with service information. These layers enhance cooperation across many teams, such as engineers, researchers, maintenance people, and decision-makers, by offering a more intuitive interaction tools for exchanging insights and making informed decisions. Thus, it is important to build an intuitive mode of interaction between users and DTs to obtain deeper discovery and understanding of the available information. Figure 10 shows the applications of service and interaction with people technologies in DT.

1) *Analytics Dashboard*: Analytics dashboards have evolved significantly to provide intuitive and data-driven insights. Analytics dashboards can integrate and visualize information to show the variation curves of monitoring data for managers and users [91]. Based on this information, DT systems can then provide visual information query tools for infrastructure operation, as well as aid in improving information usage efficiency and the formulation of maintenance strategies [92], [93]. They also incorporate advanced data visualization tools, such as interactive graphs, heatmaps, and 3D models, based on specific user needs [94]. Many studies have been performed to visualize structural information, such as measurement data, structural response prediction, and damage identification results [1], [69]. Nie et al. [92] created a BIM-based 3D visualization expression approach to allow for the intuitive expression of bridge inspection information and historical degradation data. However, due to the different formats and patterns, integrating data from several sources could be challenging.

2) *User Interfaces*: User interfaces are becoming more intuitive and responsive. User interfaces provide for the visual display of complicated data in the form of graphs, charts, and 3D models. Effective data visualization assists in developing an awareness of physical system patterns, trends, and anomalies [3]. Advanced innovations (e.g., gesture detection, voice commands, and natural language processing) provide technology capabilities that enable users to access more system information and insights. Interfaces that identify hand movements and gestures, for example, provide a more immersive and intuitive experience. This technology is especially useful

in situations where hands-free operation is preferred, such as on construction sites. Besides, user interfaces usually allow users to customize the display of information based on their preferences. Another crucial aspect for allowing quick reaction and decision-making is the provision of real-time alerts and notifications regarding significant events or changes in the physical system [95].

3) *Extended Reality*: Extended reality (XR) is highly useful in this domain since it allows users to interact with DTs in a natural way by incorporating diagrams, audio, and physical systems [96]. XR blurs the boundary between the physical and virtual worlds, resulting in engaging and immersive environments that improve the experience of the DT system. Generally, XR is classified into three types: virtual reality (VR), AR, and mixed reality (MR) [97]. VR is entirely digital content, allowing users to interact with a completely immersive environment in first-person view [98]. VR can be used for simulating scenarios, training, and visualizing data within a virtual environment. AR superimposes digital content onto the physical world, frequently employing devices like smartphones or AR glasses, thereby enhancing the real-world setting with contextual information [99]. MR combines features of both VR and AR, enabling digital objects to interact with physical objects and vice versa. In other words, MR is a digital overlay that captures spatial and temporal information from the physical world into interactive virtual elements for data visualization and analytics [4]. With XR technologies, stakeholders can fully engage with the data, acquiring more profound understandings and arriving at well-informed choices. In addition, AR and MR visualize digital content superimposed on the real world to help users perform complex tasks in transportation infrastructure, such as proactive maintenance, bridge inspection, and predictive daily operations [100], [101].

4) *Development and Challenges*: In conclusion, service and interaction with people layers are pivotal in augmenting decision-making, cooperation, and the overall usability of the DT. Analytics dashboards compile and present crucial performance metrics, trends, and insights extracted from the data. User interfaces establish a channel for users to interact with the DT, accessing its data and functionalities. User interfaces are designed to be intuitive, user-friendly, and tailored to meet the distinct requirements of various stakeholders, including engineers, maintenance personnel, and decision-makers. MR technologies meld digital information with the physical surroundings, enabling stakeholders to engage with the data within real-world contexts. These visual depictions assist decision-makers in overseeing the condition, efficiency, and safety of the physical system, while also identifying opportunities for enhancement. Through the effective integration and optimization of these components in the future, DTs can furnish invaluable insights, streamline operations, and cultivate collaboration among stakeholders, engineers, and ordinary users.

IV. APPLICATIONS

This section draws on the papers of DT for transportation infrastructure, discussing the papers in more detail and highlighting the DT concept and case studies performed. The

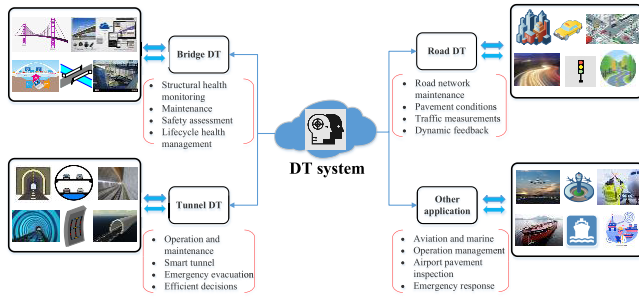


Fig. 11. Applications of DT in the area of transportation infrastructure.

sections include research papers relating to the DT concept in the domains of road, bridge, tunnel, marine, and aerospace. Figure 11 depicts the applications of DT for transportation infrastructure.

A. Bridge DT

Bridges, as a crucial component of transportation infrastructure, are prone to structural damage and material degradation due to exposure to harsh environmental conditions [3]. Structural health monitoring (SHM) systems for bridges could provide important information for maintenance, repair, and reinforcement [102]. With remarkable advancements in technologies such as 5G, IoT, AI, and cloud frameworks, DTs have emerged and received increasing attention to enable effective and economical maintenance strategies based on SHM techniques. There are two main applications of DT in the bridge domain, including lifecycle health management and safety assessment under extreme events. Table IV shows typical applications of DT in this domain. The primary aspects are discussed as follows:

In the area of lifecycle health management, to address the evolving challenges in maintaining structural integrity and ensuring the safety of infrastructure, researchers have been exploring innovative approaches to monitor and assess the health of structures. One such pioneering approach is the development and implementation of digital twin technology, which represents a significant leap forward in the realm of structural health monitoring. In this context, Dang et al. [2] put forward a framework for digital twinning in the context of structural health monitoring, leveraging cloud-based infrastructure and deep learning techniques. This framework integrated distinct submodels and facilitates the exchange of data between the physical structure, digital model, and human interventions (Figure 12). The authors also delved into the challenges inherent in deploying a cloud-based digital twin for structural health monitoring, encompassing limitations in computational resources and the prioritization of user requests. However, the scarcity of relevant data for training the models, as most available data relate to structures in a normal state, which reduces the need for transmitting and storing full data continuously. Similarly, Dan et al. [49] introduced a decision tree system designed specifically for bridges. This system utilized data from monitored traffic loads in order to perform operational condition assessments and evaluate safety risks within the transportation network. The study successfully demonstrated the feasibility of employing measured traffic

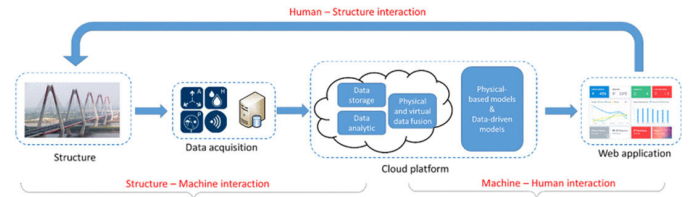


Fig. 12. Cloud system based DT model [2].

loads as an interaction task and showcased the potential of applying DT methods in the field of health monitoring. These two papers represent significant strides toward realizing the full potential of digital twinning and data analytics in enhancing the safety and longevity of infrastructure. By overcoming challenges related to data scarcity and computational limitations, these approaches offer promising solutions for the proactive management of structural health.

As extreme events like earthquakes pose a severe risk to critical infrastructures, such as bridges, innovative methodologies are essential to ensure their resilience and safety. Several researchers have proposed useful studies in safety assessment during extreme events. In the study conducted by Lin et al. [107], a collapse fragility assessment approach for long-span cable-stayed bridges during seismic activity was developed. This approach was based on the utilization of digital twin technologies and finite element models. The authors conducted tests to validate the feasibility and accuracy of the proposed method, which was found to be both feasible and accurate (Figure 13). By comparing the results obtained from the proposed digital twin-based collapse fragility assessment method with the collapse test results of the physical model of the bridge, the paper successfully demonstrated the feasibility and accuracy of the method. In parallel, Ghahari et al. [108] introduced a comprehensive framework for the digital twinning and virtual sensing of bridges, specifically focusing on damage diagnostics under earthquake conditions. Their proposed framework utilizes the time-domain model updating technique to update the FE model, thereby creating a digital twin of the bridge. This digital twin can then be employed to analyze and monitor the structural response of the bridge at various levels, including element, section, and fiber, in order to identify potential damage mechanisms. The authors demonstrated the practical applicability of their model updating technique by achieving a reasonable level of agreement between recorded and estimated bridge responses. By leveraging DT technologies, AI techniques and FE models, these studies not only enhance understanding of bridge behavior during seismic events but also pave the way for more accurate and efficient diagnostic and monitoring systems.

Overall, multiple applications have been presented for bridge performance assessment and damage detection based on FE models, AI techniques, and SHM data. DT technologies appear to be perfectly timed for use in bridge SHM systems. However, presently, these proposed frameworks operate with predetermined categories of damages and do not encompass all potential instances of failure. A specific and thorough comprehension of the interaction between information in the physical structure and the digital model is absent, indicating

TABLE IV
TYPICAL APPLICATIONS OF DT IN THE BRIDGE DOMAIN

Paper	Application	Actual Twin	Technology	Data type	Algorithm type	Interactive mode	Result
Dang <i>et al.</i> [2]	Real-time monitoring and maintenance	DT	Simulation, DL, Cloud, Visualization	Vibration	CNN	3D model, Web	Real-time monitoring and damage detection
Dan <i>et al.</i> [49]	Traffic load monitoring and bridge assessment	DT	AI, CV, Cloud	Traffic load, Traffic video	Yolo-v3, CNN	Web	working condition perception and safety warning
Jiang <i>et al.</i> [103]	Fatigue life prediction	DT	Simulation	Stress	Probabilistic multiscale crack model	N/A	Fatigue evolution throughout the bridge lifecycle
Kaewunruen <i>et al.</i> [104]	Risk-based maintenance	DS	Visualization	Crack	BIM	3D model	Vulnerability assessment and risk-based maintenance planning of bridges
Kang <i>et al.</i> [105]	Multimedia knowledge-based bridge health monitoring	DT	Simulation, AI, Visualization	Vibration, Wind, Temperature	Surrogate model	Software	Predict the optimal time point for maintenance
Mafipour <i>et al.</i> [106]	Bridge parametric modeling	DS	Visualization	Bridge point cloud	Parametric prototype model, Density based spatial clustering, Fuzzy C-Means algorithm	3D model	Generate the parametric model of bridges from segmented point clouds
Lin <i>et al.</i> [107]	Seismic collapse assessment	DT	Simulation, Visualization	Vibration, Velocity	Incremental dynamic analysis	3D model	Accurately quantify the seismic collapse resistance and identify the vulnerable components of a bridge
Ghahari <i>et al.</i> [108]	Damage diagnosis	DT	Simulation	Vibration	Sequential Bayesian model updating technique	N/A	A real-time post-earthquake damage diagnosis for bridges
Shabani and Kioumarsi [109]	Seismic assessment and strengthening	DS	Simulation, Visualization	Point cloud, Vibration	Multi-stripe analysis	N/A	Provide seismic vulnerability assessments of masonry arch bridges
Lin <i>et al.</i> [110]	Seismic performance assessment	DT	Simulation	Vibration	Monte Carlo simulation method	N/A	Provide a more accurate estimation of the service life of a long-span cable-stayed bridge

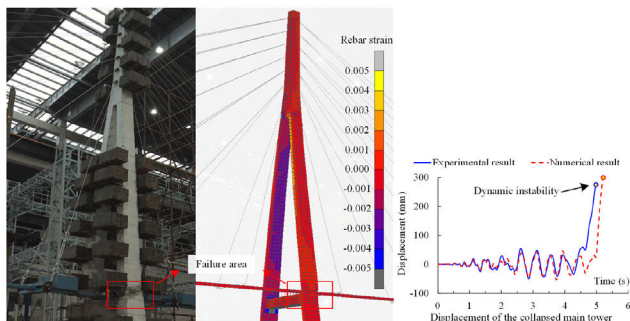


Fig. 13. Modeling and prognosis of the bridge [107].

a limitation in the current understanding of the application of DT in civil engineering. Furthermore, its utilization is primarily focused on new long-span bridges, with minimal research conducted on the application of DT to aging bridges. There is a compelling necessity for relevant organizations and departments to employ such techniques to enhance bridge management.

B. Tunnel DT

Tunnels have complicated spatial structures as compared to roadways and bridges, owing to their special characteristics

such as semi-enclosed space, buried structure, and complicated connections between components [75]. The lifecycle data of tunnels are multisource, heterogeneous, and have spatiotemporal relationships between the data, which brings many challenges to the data analysis process [111]. DTs are regarded as powerful tools, and we are beginning to see DT applications implemented in this area, with a clear focus on tunnel operation and maintenance (O&M). DTs facilitate enhanced decision-making through the integration of multi-source, heterogeneous data with spatiotemporal relationships. This technology supports the identification and analysis of potential issues, thereby enabling preemptive measures to ensure the integrity and functionality of tunnel infrastructure. Table V shows typical applications of DT in the tunnel domain. For instance, Yu et al. [75] applied semantic web technologies for multi-level data fusion and proposed a decision analysis approach using digital twins for tunnel O&M. They defined an extended COBie standard-based organization method and utilized Semantic Web technologies for fusion at different levels. This framework integrated life cycle spatio-temporal data of tunnels and analyzed potential causes and effects of abnormalities in civil structures or electromechanical equipment, providing countermeasures for O&M management. The

TABLE V
TYPICAL APPLICATIONS OF DT IN THE TUNNEL DOMAIN

Paper	Application	Actual Twin	Technology	Data type	Algorithm type	Interactive mode	Result
Yu <i>et al.</i> [75]	Operation and maintenance	DT	Visualization, Big data	Real-time sensor monitoring, manual inspection	Extended COBie standard, rule-based reasoning engine	Web	Provide reasonable and feasible countermeasures to guide and optimize O&M
Wu <i>et al.</i> [112]	Virtual-real fusion of tunnel traffic	DT	Simulation, Visualization	Surveillance video, Temperature, Wind speed, Humidity	BIM	3D model	Improve the efficiency of managing business requirements and support the accurate perception and fine management
Sanfilippo <i>et al.</i> [113]	Tunnel digitalization	DS	Visualization	Asset Information	BIM	3D model, Web	manage all the information and classification needed for O&M
Dimitrova <i>et al.</i> [74]	Tunnel diagnosis	DT	AI, Expert knowledge, Visualization	Tunnel inspection data	Decision tree model, Web Ontology Language standard	Web, software	Address the complex issues of tunnel pathology inferencing, aiding in, and matching transportation experts' expectations of decision support
Han <i>et al.</i> [37]	Pedestrian emergency evacuation	DT	Simulation	N/A	Cellular automata	3D model	Provide pedestrian emergency evacuation instructions, aiming at enhancing the overall evacuation efficiency
Hu <i>et al.</i> [111]	Defects diagnosis	DT	Visualization, Cloud, Expert knowledge	Tunnel inspection data	Semantic Web technologies, meta-standard model, BIM	3D model, Software, Web	Provide support for decision making with respect to cause detection

limitations mainly include the lack of effective mechanisms for twin data association and fusion due to the general and complex nature of existing standards like IFC and COBie, which are not fully applicable to tunnel data. Another practical exemplification can be observed in the work of Wu *et al.* [112], where a method for 3D registration and projection calculation was devised to enable the seamless integration of tunnel surveillance video into a real-time three-dimensional virtual scene (Figure 14). To enhance the effectiveness of tunnel operations and management, a management model grounded in BIM technology is employed. The research paper successfully amalgamated and merged data obtained from multiple sensing sources, resulting in the creation of a DT scene that closely resembles the actual traffic operations scene within the tunnel. Moreover, DTs not only facilitate the analysis of complex data and the identification of potential issues but also enhance the overall management and operational efficiency of tunnel infrastructure.

In summary, current studies of DT in this field are limited, although some studies have mentioned its application prospects. The absence of a proficient mechanism for the association and fusion of twin data is a contributing factor. The current standards cannot entirely apply to the spatio-temporal data association and fusion throughout the life cycle of the tunnel. Consequently, this results in a deficiency of comprehensive interaction and profound integration of twin data, which in turn impacts the real-time nature and precision of operational management decisions. Because of rapid advancements in connectivity through IoT, AI-based algorithms, such as ML and DL, possess the ability to establish a correlation

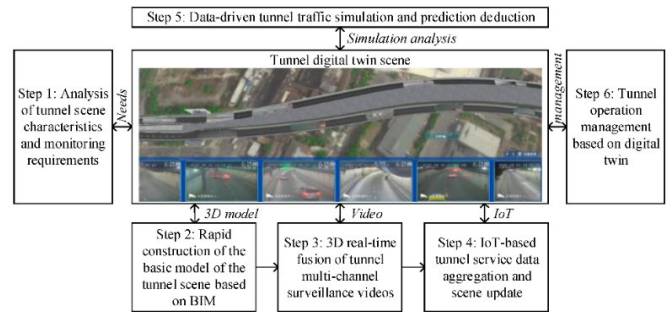


Fig. 14. The proposed tunnel DT framework [112].

between sensor data and its relevant parameters. In forthcoming times, these algorithms can be utilized for updating 3D scenes driven by data from multiple sources, simulating tunnel traffic operations driven by data, and making predictive deductions. This will result in the realization of functions that enable two-way interaction between virtual models and real models. Therefore, the development of smart tunnels using advanced technologies will help accelerate the applications for DTs in this area.

C. Road DT

In this section, we present recent representative work applying DTs in the road domain. Roads are the foundation of transportation networks. Roads provide various ways of transportation, supporting many different kinds of vehicles such as automobiles, trucks, buses, and trains. This flexibility

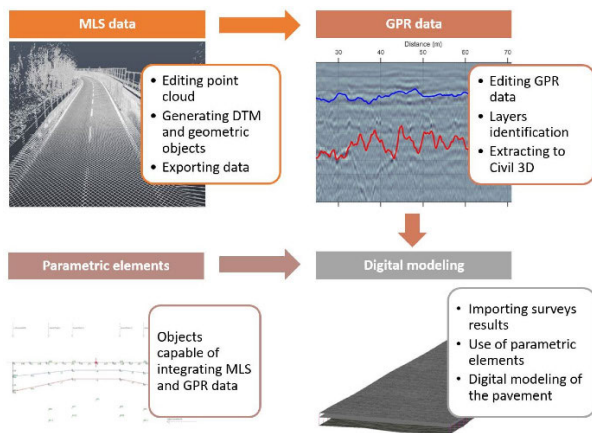


Fig. 15. Workflow for creating a digital system [116].

enables the efficient movement of individuals and goods, which contributes to a well-functioning society and economy. Most applications in the road domain focus on its monitoring and maintenance, such as pavement monitoring, road safety assessment, and railway monitoring and maintenance [114]. Table VI shows typical applications of DT in the road domain. For instance, Steyn and Broekman [115] presented a data-driven operational DT system that consists of real-time environmental data, physical data, and surface temperature data to help with road network maintenance. This system employed digital reconstruction techniques utilizing optical sensor technologies and mobile sensor platforms to construct cost-effective digital replicas of physical infrastructure. The fusion of advanced environmental monitoring data, physical data, and surface temperature data supplied management data that can support road maintenance efforts. D'Amico et al. [116] employed non-destructive technology tools like laser-scanners and ground-penetrating radar to conduct efficient surveys of pavement conditions, while maintaining the integrity of the pavement structure (Figure 15). Additionally, they utilized BIM technology to manage the dataset obtained from these surveys, and proposed a process for integrating the survey results into a BIM-oriented digital model for road pavement management, enabling continuous evaluation of pavement conditions and planning maintenance interventions. However, the model struggles to accurately capture road surface conditions and lacks detail on the deeper layers of the road's edges, indicating a need for further development and study to enhance its effectiveness.

To summarize, the most notably cases of applications in road networks is the monitoring and maintenance, such as pavement monitoring, urban planning and development, and environmental impact. By integrating real-time data, DTs can be used to monitor road conditions, detect structural problems, predict maintenance needs, and even assist in accident prevention. Advanced technologies such as ML, simulation software, and GIS are used in the road domain. These advancements facilitate the detailed assessment and proactive management of road conditions, contributing to the optimization of maintenance strategies. However, additional research and implementations are required to enhance the

suggested method and enhance the precision of the model in depicting on-site circumstances, particularly in accurately modeling specific road surface conditions and the deeper layers at the edges. Moreover, although DTs may be efficient for smaller road networks or projects, expanding them to national or regional scales may pose unexpected difficulties. We believe that this condition will change in the next few years, and scenarios such as dynamic evaluation systems of large road network resilience and sustainability appear to be promising candidates for the prospective application of DTs.

D. Other Applications

This section mainly focuses on current research using DT technologies in the aviation and marine domains. Aviation and marine also play important roles in transportation infrastructure, and they achieve an impressive level of sustainable economic and social development. However, as air and marine travels continue to grow, operation management encounters difficulties responding to increased traffic, resulting in lower operational efficiency and security concerns [122]. With the emergence of IoT and big data, research related to DT has seen significant growth in recent years. Table IV show typical applications of DT in these domains.

In recent advancements in DT within the aviation sector, various research efforts have been dedicated to enhancing airport operations and maintenance. DT utilizes cutting-edge strategies and techniques to meet the changing requirements of airport infrastructure, focusing on enhancing operational effectiveness, ensuring safety, and promoting environmental sustainability. One notable study by Keskin et al. [95] proposed a scalable BIM-based DT platform based on system engineering principles for enhancing the maintenance and operations of airport facilities. The study employed model-based system engineering (MBSE) principles with systems modeling language (SysML) for data mapping and validation. The implementation of a scalable BIM-based digital platform is anticipated to yield enhanced business value of airport infrastructures through the enhancement of operations and a reduction in initial costs for technology implementation. Another significant contribution by Gui et al. [123] proposed a cloud-edge-terminal-based distributed framework for airport pavement inspection tasks using the robotic system to automate the inspection process, enhance coverage, and reduce manpower and time costs (Figure 16). The suggested system comprised three primary components: Terminal (a data collection system based on a wheeled robot), Edge (a system for remote monitoring and data analysis), and Cloud (a shared database center for inspection data and knowledge). The robotic system ensured comprehensive coverage and delivered pavement inspection results with high precision, while requiring less time and labor costs in comparison to the conventional manual approach. The paper highlights the necessity for human intervention in the analysis and confirmation of data, alongside pointing out the ambition to enhance algorithm efficiency and the integration of multiple robots for comprehensive inspections across vast areas. Current studies promise to enhance operational efficiency, reduce costs, and improve

TABLE VI
TYPICAL APPLICATIONS OF DT IN THE ROAD DOMAIN

Paper	Application	Actual Twin	Technology	Data type	Algorithm type	Interactive mode	Result
Steyn and Broekman [115]	Road maintenance	DT	Visualization, Cloud, AI	Road surface temperature	Neural network	Web	Provide management data that can assist in the maintenance of roads
D'Amico <i>et al.</i> [116]	Road pavements management	DS	Visualization	Point cloud, Ground penetrating radar data	BIM, Laser-scanner, Ground penetrating radar	3D model	Evaluate continuously the pavement conditions and programming maintenance interventions
Cao <i>et al.</i> [117]	Pavement Crack Detection	DT	Cloud, AI, Visualization	Video, Point cloud	3D scanning technology, Crack segmentation algorithm	3D model, Software	Accomplish crack detection satisfactory results in complex pavement background
Barisic <i>et al.</i> [118]	Asphalt road assessment	DT	Visualization	Temperature	Thermal modelling	N/A	Provide the thermal state of asphalt roads
Sotiriadis <i>et al.</i> [119]	Pavement monitoring	DT	Simulation, ML, Visualization	Strain	Artificial Neural Network	3D model	Monitor and optimize assets and make informed decisions
Kampczyk and Dybel [120]	Turnout assessment	DS	Visualization	Temperature	N/A	Software	Provide information on the technical condition of a turnout
Sierra <i>et al.</i> [121]	Pavement crack detection	DT	DL, Visualization	Crack image	U-Net, CNN	N/A	Detect damage on the pavement and enhance efficiency of maintaining pavement

TABLE VII
OTHER TYPICAL APPLICATIONS OF DT

Paper	Application	Actual Twin	Technology	Data type	Algorithm type	Interactive mode	Result
Aviation area							
Keskin <i>et al.</i> [95]	Airport asset management	DT	Cloud, Big data, Visualization	Airport asset data	BIM, Model-based system engineering, Systems modeling language	3D model, Web	Optimize the collection, integration, management, and utilization of airport asset data
Gui <i>et al.</i> [123]	Airport runway inspection	DT	Cloud, visualization	Ground penetrating radar data	Region-convolutional neural networks	3D model, Web, Software	Provide high-precision pavement inspection results with less time and labor costs
Shao <i>et al.</i> [124]	Fire risk analysis	DS	Simulation, Visualization	Temperature, Wind speed	Bayesian network, Probability correction model	3D model, VR	Provide scientific guidance for the airport's aviation safety management
Marine area							
Li <i>et al.</i> [125]	Port safety evaluation	DT	Optimization, Visualization	Vibration, Strain, Stress, Settlement	Floating stereoscopic display technology	VR	Offer a new immersive and interactive mode for port structure safety monitoring and analysis
Wu <i>et al.</i> [126]	Safety management of inland waterway	DT	Visualization	Video, Point cloud	BIM, Drone tilt photography	AR, VR	Reduce the pressure on managers to understand real-time video footage and improve the work efficiency
Yigit <i>et al.</i> [127]	Port maintenance and management	DS	Simulation, Optimization, cloud	Inspection data, Ship parameter data	TwinPort architecture	3D model	Offer a comprehensive port management system for smart seaports

infrastructure management. However, they also underscore the ongoing need for human expertise in data analysis and the potential for further advancements in algorithmic efficiency and collaborative robotic systems to meet the challenges of large-scale airport maintenance and inspection tasks.

In the rapidly advancing field of DT technology, researchers are continuously exploring new frontiers for its application in the marine sector. DT technology is being harnessed to enhance the safety and efficiency of maritime operations. For example, Li *et al.* [125] proposed a method to improve the visualization of multi-source data fusion for creating a digital

twin model of the port structure and auxiliary monitoring system based on the theory of virtual-real mapping port digital twin modeling. In details, this study developed floating stereoscopic display equipment and achieved visual display and natural interaction of the digital twin for the safety monitoring of the port structure, which is a new approach for maintaining port structure safety. Wu *et al.* [126] established a DT framework for the safety management of inland waterways based on 3D video fusion to improve the efficiency of daily monitoring, data collection, and emergency response (Figure 17). It combined various IoT sensing data sources

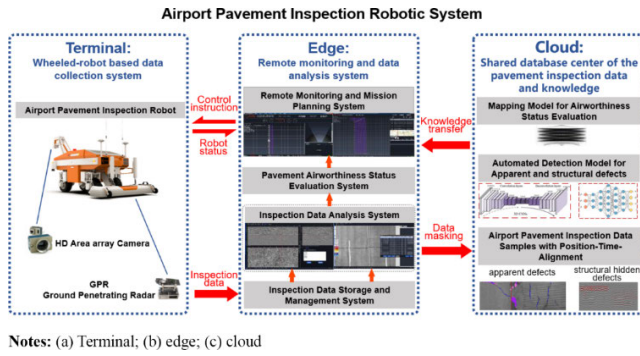


Fig. 16. Overall system architecture of our cloud-edge-terminal-based airport pavement inspection robotic system [123].

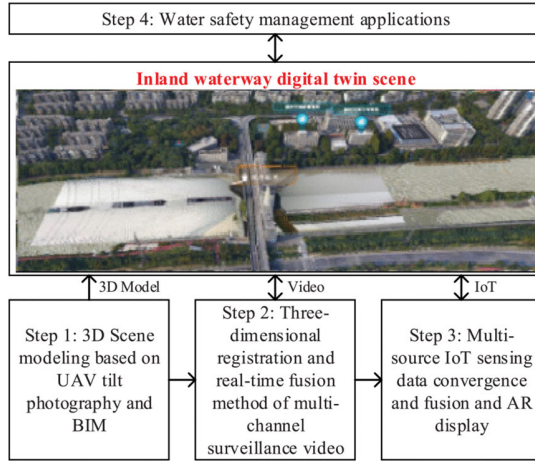


Fig. 17. DT construction and safety management method of inland waterways based on 3D video fusion [126].

to create a digital replica of inland waterways, improving our grasp on current waterway situations. This research filled the gap in studies focusing on digital safety oversight for inland waterways and offered valuable insights and strategies for their safety management. Through the innovative use of multi-source data fusion and virtual-real mapping, researchers are developing DTs that offer unprecedented visualization and interaction capabilities. These models are crucial for the safety monitoring and management of port and waterways structures.

To summarize this section, most of the aforementioned studies for transportation infrastructure explicitly employ DT technologies. SHM and O&M management techniques are increasingly economically viable in these areas. In contrast, DTs in the bridge domain have established relatively concrete tools, especially in the aspect of SHM. However, the development of DTs for transportation infrastructure is still in its early stage, and full integration of DTs with emerging technologies (e.g., IoT, AI, Big Data, etc.) is necessary. One could also note that the scale of the application is small currently for transportation infrastructure, which indicates complexities and challenges in implementing DTs in practice.

V. CHALLENGES AND OPEN RESEARCH

A. Challenges

In this section, we will summarize the challenges of DT technology applied to the transportation infrastructures presented above. Currently, DT has been increasingly adopted

in transportation infrastructure; however, most studies are still immature due to the complexity of practical problems in reality and the lack of technologies to achieve high accuracy and real-time performance of DT. In particular, this section covers several critical challenges encountered in the development and application of DT for transportation infrastructure, as discussed below.

1) *Model Complexity*: A high-fidelity model is highly desired for DTs, but they are usually very complex to build and time-consuming for execution. The design blueprint of transportation infrastructure and its corresponding runtime state in the real physical world are often different. This is because the construction and management processes of real infrastructure are affected by many uncertainties, and the causal relationship is extremely complex. Hence, the DT system not only depends on the information and model in the design phase as a guide, but also needs real-time information that is transformed from a physical system to update the constructed system. On the other hand, high-fidelity models can more accurately reflect the real state of the related system. However, due to its time-consuming computation, it can only simulate small-scale structures. Thus, it is necessary to establish an effective strategy that achieves or at least balances the high-fidelity model and high-efficiency simulation.

2) *Model Updating*: Civil infrastructures are subjected to material aging, structural deterioration, and even natural hazards such as earthquakes, floods, and typhoons. The model updating method provides a solution for accurate prediction of the structural response and structural damage detection. However, most engineering systems contain limited information and monitoring data, which leads to poor prediction performance of the established data-driven model. Another issue is data quality, which can be influenced by factors such as ambient temperature, noise, and measurement error in addition to infrastructure. As a result, it remains a challenge to effectively use limited information or incomplete data for structural response modeling and prediction. Furthermore, several studies have demonstrated that the physics-based model could improve model accuracy with inadequate data [87], [108]. However, reducing the bias of model structure, which is caused by inaccurate or inadequate knowledge of fundamental physical principles, may still be challenging.

3) *Control and Intervention*: The service layer occupies a significant position in the DT framework, which is also the goal of building a DT system. Many studies have proved that DT systems can assess the health status of physical infrastructures and make decisions based on the available information. However, it is not sufficient to control the DT system economically and effectively at present. It is vital to verify the practical applicability of the interventions recommended by the DT to the actual physical system. In addition, in situations where immediate intervention is required, the DT must provide prompt and precise information. Any delays or inaccuracies could lead to missed chances for efficacious intervention. Thus, DT systems should contribute to long-range, full, and accurate service systems for transportation infrastructure, and offer smart services for managers to make decisions and solutions sustainably.

4) *Integrated Infrastructure*: Integrating infrastructure involves the seamless connection and interoperability of structural components, systems, and data to establish a comprehensive system within the DT. Integrating the information from all layers of DT systems is challenging since most experts and engineers have limited knowledge of a given problem. Most DT studies are conducted through various types of software that are designed specifically for one knowledge domain, which inevitably leads to an inaccurate DT system. In addition, the development of systemic DTs is restricted by the lack of agreement on different criteria, technologies, and procedures for various domains. Integrating data from several domains into DT results in a massive amount of information, which is usually associated with high costs. This impedes the development of the systematization of DTs and leads to a widening gap between DT systems from different areas.

5) *Interpretation and Building Trust*: Another challenge is how to make the data generated by the DT system easier for managers to understand, so as to better show the advantages of DT technology. DT systems, in particular, should support improved decision making by offering a platform for performing and evaluating physical infrastructure. However, most researchers are engaged in developing DT systems instead of delivering DTs to the public. Because of familiarity, many engineers and managers find their current traditional decision-making procedures (e.g., empirical formulas and judgments) for infrastructure management more practical. Thus, it is considered that DTs can simulate and assess physical systems for academics and experts, but nontechnical managers and ordinary users assert that they cannot be referred to as credible and persuasive evidence owing to complicated and unintuitive systems and algorithms. In other words, more convincing evidence is required to show the potential and public value of DT systems.

6) *Application to aging structures*: The use of DT technology in monitoring and maintaining aging transportation infrastructures shows potential for improving their longevity, safety, and performance. Unlike newly built structures, aging structures face many challenges that hinder the application of DT. For example, over time, the degradation of materials such as concrete, steel, and wood can pose challenges in accurately assessing their state. Because the original baseline data cannot be obtained for aging structures, leaving engineers without a reference point. On the other hand, the deficiency of comprehensive data regarding aging transportation infrastructures can hinder the complete exploitation of the advantages presented by DT technologies. Erroneous or incomplete data can exacerbate inaccuracies within the DT representation of the physical infrastructure. Lastly, stakeholders may be reluctant to invest in advanced monitoring systems due to the perceived limited lifespan of older structures, instead choosing to allocate funds to newer structures.

B. Open Research

As the DT system of transportation infrastructure is a comprehensive subject involving many fields, the development of DT herein is still in its infancy. In the future, DTs will

possibly embrace four research directions: multi-scale/fidelity modeling, what-if simulation, system engineering perspectives, and other opportunities. Figure 18 shows a roadmap for future investigations and innovations of DT in transportation infrastructure.

1) *Multi-scale/fidelity Modeling*: The virtual system of DT is the true reflection of the real system, which is the foundation for the DT system. On the one hand, a real structure has many physical properties (e.g., material stiffness, mass, fatigue strength, and temperature), and each of these physical properties has its own specific model, such as the fatigue damage model, structural static and dynamic models, and aerodynamic model. On the other hand, it is necessary to consider multi-rate simulation within a DT system for accurate simulation. For example, some parts of a DT tunnel system have high dynamic properties, e.g., rails and trains, so they need high-rate simulation to capture dynamic properties. But for other parts, like soil movement, they have a large volume but mostly static or pseudo-static performance, which indicates low-rate simulation is preferred. Integration of these systems with different parametric values, frequencies, and time scales into the DT system should be adequately studied and designed. Meanwhile, computation cost and time should also be considered for the real-time reflection of the DT system.

2) *Advanced Data Fusion and DL for Model Updating*: Advanced data fusion techniques and DL algorithms aims to address the challenge of limited and incomplete data by efficiently combining diverse data sources, such as sensor data, satellite imagery, and drone footage, to enhance the accuracy and reliability of structural health monitoring and prediction models. Furthermore, the development of physics-informed DL models, which incorporate fundamental physical principles, offers a promising avenue to reduce biases caused by inadequate data and improve the predictive performance of DT systems. By advancing these techniques, researchers can significantly improve the capability of DTs to provide accurate, real-time updates on the structural integrity and performance of transportation infrastructures, enabling more effective maintenance and management strategies.

3) *What-if Simulation*: Integrating intelligent analytic technology with DT systems assesses and recommends the present structure, increasing the speed and precision of decision-making for managers and users. For example, Dan et al. [49] proposed a DT system for bridges that uses monitoring traffic loads to analyze operational status and safety risks in the transportation network. In fact, although DTs can offer valuable insights, human expertise remains pivotal in the decision-making process. Typically, managers make ultimate decisions by considering real-time information and recommendations derived from both simulated and forecasted results. In other words, the DT should serve as a complement to human judgment, rather than a substitute for it. The next generation of the DT system should analyze many scenarios of strategies (e.g., working time, labor force, and capital cost) to offer comprehensive suggestions for the next action. Then, it allows decision-makers to effectively handle unknown situations and decide on the best available option based on the comparison results.

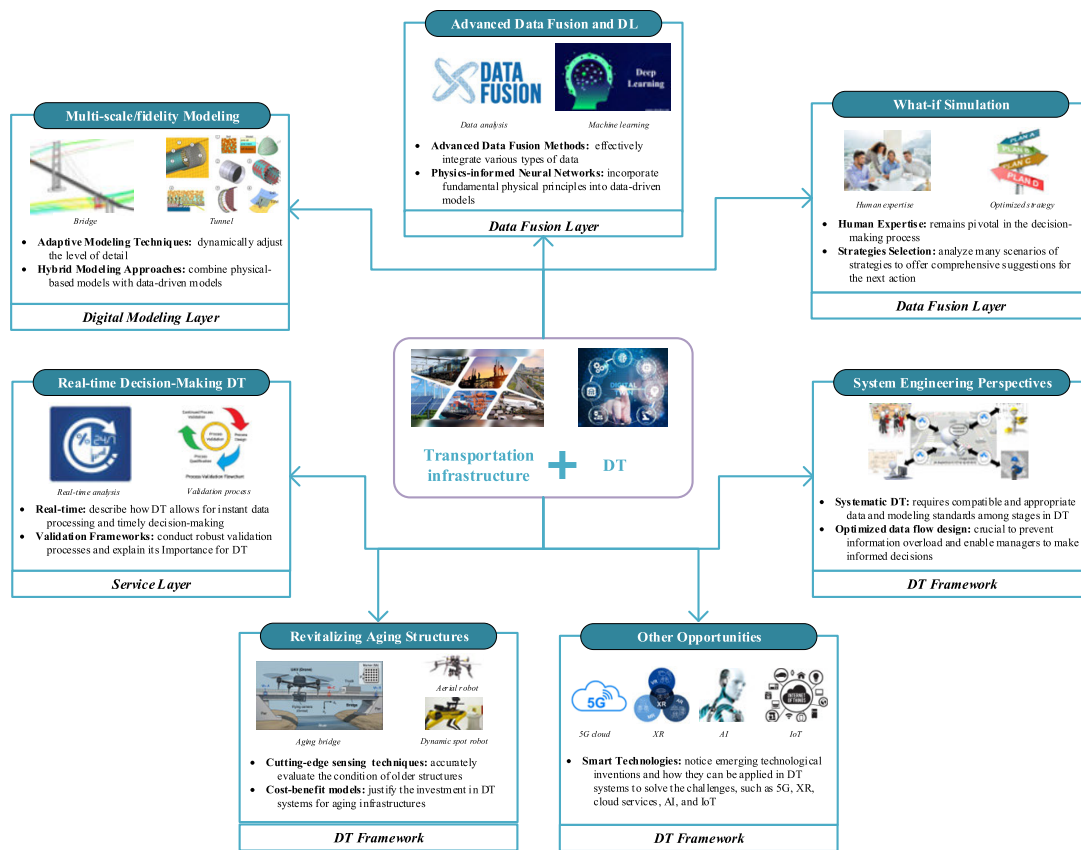


Fig. 18. Future directions of DT in transportation infrastructure.

4) *Real-time Decision-Making DT*: The development of the real-time decision-making features of DTs is also prioritized in future research. This involves creating sophisticated DT systems capable of delivering prompt and accurate data for swift interventions, crucial in emergency scenarios or for proactive maintenance. Incorporating advanced analytics, AI, and other algorithms can significantly enhance the DT's ability to analyze vast amounts of data in real time, offering actionable insights for immediate control measures. Moreover, establishing comprehensive validation and verification frameworks is essential to ensuring the reliability and applicability of DT recommendations, fostering trust among infrastructure managers and stakeholders. By focusing on these areas, DT technology can evolve into a more dynamic and effective tool for infrastructure management, enabling smarter, data-driven decisions that ensure safety, efficiency, and sustainability.

5) *System Engineering Perspectives*: System engineering is a holistic and integrated approach that involves the complete lifecycle of a system, ranging from its initial conceptualization to its operational phase. The development of a systematic DT requires compatible and appropriate data and modeling standards among stages of the same area, even within different areas. It encourages cooperation among diverse fields of expertise and stakeholders engaged in the creation and functioning of a system [11]. For instance, a systematic tunnel DT could include structural monitoring and evaluation system, groundwater information system, temperature system, passenger and train system, etc. Data from various sources usually

exhibit variations in formats and arrangements. Guaranteeing conformity to data standards and facilitating interoperability across these sources is essential to construct a comprehensive DT of the real-world system. Meanwhile, the flow of data and information should be properly designed and optimized based on data interdependencies of different subsystems to avoid an information explosion. Managers from different areas can then access the monitoring data and simulated results to make informed decisions.

6) *Revitalizing Aging Structures with DT*: Addressing the unique challenges of aging transportation infrastructures through DT technology is a critical future research direction. This entails the development of specialized monitoring and assessment tools that can accurately evaluate the condition of older structures, despite the absence of original baseline data. Emphasis should be placed on leveraging cutting-edge techniques such as advanced sensing technologies, data imputation, and reconstruction methods to overcome the limitations posed by data gaps and material degradation. Furthermore, creating cost-benefit models to justify the investment in DT systems for aging infrastructures is essential, highlighting the potential for significant improvements in safety, performance, and longevity. By tailoring DT approaches to the specific needs of aging structures, researchers can unlock new possibilities for extending their service life and enhancing infrastructure resilience.

7) *Other Opportunities*: With the rapid growth of advanced smart technologies (e.g., 5G, XR, cloud services, AI, and IoT),

we should notice emerging technological inventions and how they can be applied in DT systems to solve the aforementioned challenges. For example, 5G technology could be used to address the challenges of latency, scale, and complexity at the transmission layer. Cloud services could be used as a platform for data storage and processing at a much lower cost, making the integration of data analytics and AI algorithms faster and more feasible. XR sits within services used for providing social services and interacting with managers and citizens in 3D space. In other words, these advanced technologies offer new opportunities, allowing further model fidelity and simulation efficiency for DT systems.

VI. CONCLUSION

As an emerging technology, digital twin (DT) technology has developed rapidly in recent years, and it is regarded as an important basis for the digital transformation process. DT has been developed and applied in many fields, such as manufacturing, healthcare, and smart city. This paper presents a review of DTs for transportation infrastructure based on the underlying technologies, implementations, challenges, and future directions. The following conclusions are obtained:

Based on advanced techniques presented in the DT studies, the key enabling techniques and methods for building the virtual systems of DT are summarized, such as mechanical modelling, geographic information system (GIS), city information modeling (CIM), machine/deep learning, and extended reality (XR). The key enabling techniques and their advantages and limitations of existing investigations are presented. The key enabling techniques can be used as a reference for the inspection, monitoring, maintenance, and management processes of transportation infrastructure in the future advances of DT.

DTs have great advantages in structural health monitoring (SHM), operation and maintenance (O&M), and lifecycle management for transportation infrastructure. In particular, the primary areas of interest are in the bridge domain, such as bridge inspection, monitoring, and maintenance. Compared with the research outputs on DT in other fields of transportation infrastructure, the number of papers in the bridge field is significantly higher. A few studies are in the tunnel and road domains, which primarily use DT technologies to solve specific management and maintenance problems. In terms of aviation and marine domains, as air and marine travels continue to grow, most of the current cases of DT are in operation management, responding to lower operational efficiency and security concerns.

The challenges facing the implementation of DTs for transportation infrastructure are related to model complexity, model updating, control and intervention, integrated infrastructure, interpretation and building trust with people, and application to aging structures. From the research challenges identified in this paper, open research directions for the future are then provided, including multi-scale/fidelity modeling, advanced data fusion and DL for model updating, what-if simulation, real-time decision-making DT, system engineering perspectives, revitalizing aging structures with DT, and other opportunities. Moreover, a more scalable and unified DT framework would

greatly increase the value of DT research. Finally, although the DT technology is still in its infancy, this review provides a foundation for further work.

REFERENCES

- [1] M. Deng, C. C. Menassa, and V. R. Kamat, "From BIM to digital twins: A systematic review of the evolution of intelligent building representations in the AEC-FM industry," *J. Inf. Technol. Construct.*, vol. 26, pp. 58–83, Jan. 2021.
- [2] H. Dang, M. Tatipamula, and H. X. Nguyen, "Cloud-based digital twinning for structural health monitoring using deep learning," *IEEE Trans. Ind. Informat.*, vol. 18, no. 6, pp. 3820–3830, Sep. 2022.
- [3] M. M. Futai, T. N. Bittencourt, H. Carvalho, and D. M. Ribeiro, "Challenges in the application of digital transformation to inspection and maintenance of bridges," *Struct. Infrastruct. Eng.*, vol. 18, nos. 10–11, pp. 1581–1600, 2022.
- [4] N. Mohammadi and J. E. Taylor, "Smart city digital twins," in *Proc. IEEE Symp. Comput. Intell. (SSCI)*, Nov. 2017, pp. 1–5.
- [5] A. Fuller, Z. Fan, C. Day, and C. Barlow, "Digital twin: Enabling technologies, challenges and open research," *IEEE Access*, vol. 8, pp. 108952–108971, 2020.
- [6] Z. Liu, N. Meyendorf, and N. Mrad, "The role of data fusion in predictive maintenance using digital twin," in *Proc. AIP Conf.*, 2018, Art. no. 020023.
- [7] Z. Huang, Y. Shen, J. Li, M. Fey, and C. Brecher, "A survey on AI-driven digital twins in industry 4.0: Smart manufacturing and advanced robotics," *Sensors*, vol. 21, no. 19, p. 6340, Sep. 2021.
- [8] T. Bergs, S. Gierlings, T. Auerbach, A. Klink, D. Schraknepper, and T. Augspurger, "The concept of digital twin and digital shadow in manufacturing," *Proc. CIRP*, vol. 101, pp. 81–84, Jan. 2021.
- [9] S. M. E. Sepasgozar, "Differentiating digital twin from digital shadow: Elucidating a paradigm shift to expedite a smart, sustainable built environment," *Buildings*, vol. 11, no. 4, p. 151, Apr. 2021.
- [10] M. Shahzad, M. T. Shafiq, D. Douglas, and M. Kassem, "Digital twins in built environments: An investigation of the characteristics, applications, and challenges," *Buildings*, vol. 12, no. 2, p. 120, Jan. 2022.
- [11] G. Mylonas et al., "Digital twins from smart manufacturing to smart cities: A survey," *IEEE Access*, vol. 9, pp. 143222–143249, 2021.
- [12] Q. Lu et al., "Developing a digital twin at building and city levels: Case study of West Cambridge campus," *J. Manag. Eng.*, vol. 36, no. 3, 2020, Art. no. 05020004.
- [13] F. Tao, J. Cheng, Q. Qi, M. Zhang, H. Zhang, and F. Sui, "Digital twin-driven product design, manufacturing and service with big data," *Int. J. Adv. Manuf. Technol.*, vol. 94, nos. 9–12, pp. 3563–3576, 2017.
- [14] J. Leng, H. Zhang, D. Yan, Q. Liu, X. Chen, and D. Zhang, "Digital twin-driven manufacturing cyber-physical system for parallel controlling of smart workshop," *J. Ambient Intell. Humanized Comput.*, vol. 10, no. 3, pp. 1155–1166, Mar. 2019.
- [15] Q. Qi, F. Tao, Y. Zuo, and D. Zhao, "Digital twin service towards smart manufacturing," *Proc. CIRP*, vol. 72, pp. 237–242, Jan. 2018.
- [16] Q. Qi et al., "Enabling technologies and tools for digital twin," *J. Manuf. Syst.*, vol. 58, pp. 3–21, Jan. 2021.
- [17] G. L. Knapp et al., "Building blocks for a digital twin of additive manufacturing," *Acta Mater.*, vol. 135, pp. 390–399, Aug. 2017.
- [18] A. J. H. Redelinghuys, A. H. Basson, and K. Kruger, "A six-layer architecture for the digital twin: A manufacturing case study implementation," *J. Intell. Manuf.*, vol. 31, no. 6, pp. 1383–1402, Aug. 2020.
- [19] S. Agostinelli, F. Cumo, G. Guidi, and C. Tomazzoli, "Cyber-physical systems improving building energy management: Digital twin and artificial intelligence," *Energies*, vol. 14, no. 8, p. 2338, Apr. 2021.
- [20] R. Wagner, B. Schleich, B. Haefner, A. Kuhnle, S. Wartack, and G. Lanza, "Challenges and potentials of digital twins and industry 4.0 in product design and production for high performance products," *Proc. CIRP*, vol. 84, pp. 88–93, Jan. 2019.
- [21] A. Francisco, N. Mohammadi, and J. E. Taylor, "Smart city digital twin-enabled energy management: Toward real-time urban building energy benchmarking," *J. Manag. Eng.*, vol. 36, no. 2, 2020, Art. no. 04019045.
- [22] S. Gahlot, S. R. N. Reddy, and D. Kumar, "Review of smart health monitoring approaches with survey analysis and proposed framework," *IEEE Internet Things J.*, vol. 6, no. 2, pp. 2116–2127, Apr. 2019.
- [23] H. Laaki, Y. Miche, and K. Tammi, "Prototyping a digital twin for real time remote control over mobile networks: Application of remote surgery," *IEEE Access*, vol. 7, pp. 20325–20336, 2019.

- [24] Y. Liu et al., "A novel cloud-based framework for the elderly healthcare services using digital twin," *IEEE Access*, vol. 7, pp. 49088–49101, 2019.
- [25] S. Yang et al., "Scalable digital neuromorphic architecture for large-scale biophysically meaningful neural network with multi-compartment neurons," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 1, pp. 148–162, Jan. 2020.
- [26] S. Yang et al., "BiCoSS: Toward large-scale cognition brain with multigranular neuromorphic architecture," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 7, pp. 2801–2815, Jul. 2022.
- [27] J. Wu, Y. Yang, X. Cheng, H. Zuo, and Z. Cheng, "The development of digital twin technology review," in *Proc. Chin. Autom. Congr. (CAC)*, Nov. 2020, pp. 4901–4906.
- [28] J. Guo and Z. Lv, "Application of Digital Twins in multiple fields," *Multimedia Tools Appl.*, vol. 81, no. 19, pp. 26941–26967, 2022.
- [29] G. B. Ozturk, "Digital twin research in the AECO-FM industry," *J. Building Eng.*, vol. 40, Aug. 2021, Art. no. 102730.
- [30] Y. Chen, "Integrated and intelligent manufacturing: Perspectives and enablers," *Engineering*, vol. 3, no. 5, pp. 588–595, Oct. 2017.
- [31] A. Madni, C. Madni, and S. Lucero, "Leveraging digital twin technology in model-based systems engineering," *Systems*, vol. 7, no. 1, p. 7, 2019.
- [32] Y. Zheng, S. Yang, and H. Cheng, "An application framework of digital twin and its case study," *J. Ambient Intell. Humanized Comput.*, vol. 10, no. 3, pp. 1141–1153, Mar. 2019.
- [33] J. A. Erkoyuncu, P. Butala, and R. Roy, "Digital twins: Understanding the added value of integrated models for through-life engineering services," *Proc. Manuf.*, vol. 16, pp. 139–146, Jan. 2018.
- [34] S. P. Ramu et al., "Federated learning enabled digital twins for smart cities: Concepts, recent advances, and future directions," *Sustain. Cities Soc.*, vol. 79, Apr. 2022, Art. no. 103663.
- [35] G. Yu, S. Zhang, M. Hu, and Y. K. Wang, "Prediction of highway tunnel pavement performance based on digital twin and multiple time series stacking," *Adv. Civil Eng.*, vol. 2020, Dec. 2020, Art. no. 8824135.
- [36] J. Kim and S.-A. Kim, "Lifespan prediction technique for digital twin-based noise barrier tunnels," *Sustainability*, vol. 12, no. 7, p. 2940, Apr. 2020.
- [37] T. Han, J. Zhao, and W. Li, "Smart-guided pedestrian emergency evacuation in slender-shape infrastructure with digital twin simulations," *Sustainability*, vol. 12, no. 22, p. 9701, Nov. 2020.
- [38] K. Reifsnider and P. Majumdar, "Multiphysics stimulated simulation digital twin methods for fleet management," in *Proc. 54th AIAA/ASME/ASCE/AHS/ASC Struct., Struct. Dyn., Mater. Conf.*, 2014, p. 11.
- [39] S. H. Khajavi, N. H. Motlagh, A. Jaribion, L. C. Werner, and J. Holmström, "Digital twin: Vision, benefits, boundaries, and creation for buildings," *IEEE Access*, vol. 7, pp. 147406–147419, 2019.
- [40] M. Mashaly, "Connecting the twins: A review on digital twin technology & its networking requirements," *Proc. Comput. Sci.*, vol. 184, pp. 299–305, Jan. 2021.
- [41] T. Bauernhansl, S. Hartleif, and T. Felix, "The digital shadow of production—A concept for the effective and efficient information supply in dynamic industrial environments," *Proc. CIRP*, vol. 72, pp. 69–74, Jan. 2018.
- [42] C. Chen, "Science mapping: A systematic review of the literature," *J. Data Inf. Sci.*, vol. 2, no. 2, pp. 1–40, May 2017.
- [43] N. J. Eck and L. Waltman, *VOSviewer Manual*. Accessed: Jan. 9, 2023. [Online]. Available: https://www.vosviewer.com/documentation/Manual_VOSviewer_1.6.16.pdf
- [44] F. Tao et al., "Five-dimension digital twin model and its ten applications," *Comput. Integr. Manuf. Syst.*, vol. 25, pp. 1–18, Jan. 2019.
- [45] Z. Z. Hu, P. L. Tian, S. W. Li, and J. P. Zhang, "BIM-based integrated delivery technologies for intelligent MEP management in the operation and maintenance phase," *Adv. Eng. Softw.*, vol. 115, pp. 1–16, Jan. 2018.
- [46] H. Boyes and T. Watson, "Digital twins: An analysis framework and open issues," *Comput. Ind.*, vol. 143, Dec. 2022, Art. no. 103763.
- [47] K. Kim, Y. K. Cho, and K. Kim, "BIM-based decision-making framework for scaffolding planning," *J. Manag. Eng.*, vol. 34, no. 6, 2018, Art. no. 04018046.
- [48] B. N. Silva, M. Khan, and K. Han, "Integration of big data analytics embedded smart city architecture with RESTful web of things for efficient service provision and energy management," *Future Gener. Comput. Syst.*, vol. 107, pp. 975–987, Jun. 2020.
- [49] D. Dan, Y. Ying, and L. Ge, "Digital twin system of bridges group based on machine vision fusion monitoring of bridge traffic load," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 11, pp. 22190–22205, Nov. 2022.
- [50] F. Tao, M. Zhang, and A. Y. C. Nee, "Five-dimension digital twin modeling and its key technologies," in *Digital Twin Driven Smart Manufacturing*, F. Tao, M. Zhang and A. Y. C. Nee, Eds. New York, NY, USA: Academic, 2019, ch. 3, pp. 63–81.
- [51] L. Xu, A. Francisco, J. E. Taylor, and N. Mohammadi, "Urban energy data visualization and management: Evaluating community-scale eco-feedback approaches," *J. Manag. Eng.*, vol. 37, no. 2, Mar. 2021, Art. no. 04020111.
- [52] X. Chang, H. Wang, Y. Zhang, F. Wang, and Z. Li, "Bayesian prediction of tunnel convergence combining empirical model and relevance vector machine," *Measurement*, vol. 188, Jan. 2022, Art. no. 110621.
- [53] X. Chang, H. Wang, and Y. Zhang, "Back analysis of rock mass parameters in tunnel engineering using machine learning techniques," *Comput. Geotechnics*, vol. 163, Nov. 2023, Art. no. 105738.
- [54] Y. Zhang and W. Lin, "Computer-vision-based differential remeshing for updating the geometry of finite element model," *Comput.-Aided Civil Infrastruct. Eng.*, vol. 37, no. 2, pp. 185–203, 2021.
- [55] Z. Liu, C. Yuan, Z. Sun, and C. Cao, "Digital twins-based impact response prediction of prestressed steel structure," *Sensors*, vol. 22, no. 4, p. 1647, Feb. 2022.
- [56] Y. Xing, Q. Liang, G. Wang, X. Ming, and X. Xia, "City-scale hydrodynamic modelling of urban flash floods: The issues of scale and resolution," *Natural Hazards*, vol. 96, no. 1, pp. 473–496, Mar. 2019.
- [57] M. El-Mekawy, A. Östman, and I. Hijazi, "A unified building model for 3D urban GIS," *ISPRS Int. J. Geo-Inf.*, vol. 1, no. 2, pp. 120–145, 2012.
- [58] S. Shirovzhan, W. Tan, and S. M. E. Sepasgozar, "Digital twin and CyberGIS for improving connectivity and measuring the impact of infrastructure construction planning in smart cities," *ISPRS Int. J. Geo-Inf.*, vol. 9, no. 4, p. 240, Apr. 2020.
- [59] A. H. Hor, A. Jadidi, and G. Sohn, "BIM-GIS integrated geospatial information model using semantic web and RDF graphs," *ISPRS Ann. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 3, no. 4, pp. 73–79, 2016.
- [60] M. Breunig et al., "Geospatial data management research: Progress and future directions," *ISPRS Int. J. Geo-Inf.*, vol. 9, no. 2, p. 95, Feb. 2020.
- [61] Y. Liu, A. Padmanabhan, and S. Wang, "CyberGIS gateway for enabling data-rich geospatial research and education," *Concurrency Comput., Pract. Exper.*, vol. 27, no. 2, pp. 395–407, 2015.
- [62] S. Shirovzhan, S. Lim, J. Trinder, H. Li, and S. M. E. Sepasgozar, "Data mining for recognition of spatial distribution patterns of building heights using airborne LiDAR data," *Adv. Eng. Informat.*, vol. 43, Jan. 2020, Art. no. 101033.
- [63] X. Xu, L. Ding, H. Luo, and L. Ma, "From building information modeling to city information modeling," *J. Inf. Technol. Construct.*, vol. 19, pp. 292–307, Jan. 2014.
- [64] H. Omrany, A. Ghaffarianhoseini, A. Ghaffarianhoseini, and D. J. Clements-Croome, "The uptake of city information modelling (CIM): A comprehensive review of current implementations, challenges and future outlook," *Smart Sustain. Built Environ.*, vol. 12, no. 5, pp. 1090–1116, 2022.
- [65] M. Deng, V. J. L. Gan, Y. Tan, A. Joneja, and J. C. P. Cheng, "Automatic generation of fabrication drawings for Façade mullions and transoms through BIM models," *Adv. Eng. Informat.*, vol. 42, Oct. 2019, Art. no. 100964.
- [66] H. Sun, Z. Liu, and Z. Lv, "Research on intelligent dispatching system management platform for construction projects based on digital Twin and BIM technology," *Adv. Civil Eng.*, vol. 2022, Jan. 2022, Art. no. 8273451.
- [67] S. Tang, D. R. Shelden, C. M. Eastman, P. Pishdad-Bozorgi, and X. Gao, "A review of building information modeling (BIM) and the Internet of Things (IoT) devices integration: Present status and future trends," *Automat. Construct.*, vol. 101, pp. 127–139, May 2019.
- [68] J. Gubbi, R. Buyya, S. Marusic, and M. Palaniswami, "Internet of Things (IoT): A vision, architectural elements, and future directions," *Future Gener. Comput. Syst.*, vol. 29, no. 7, pp. 1645–1660, 2013.
- [69] C. Z. Li, F. Xue, X. Li, J. Hong, and G. Q. Shen, "An Internet of Things-enabled BIM platform for on-site assembly services in prefabricated construction," *Automat. Construct.*, vol. 89, pp. 146–161, May 2018.
- [70] G. White, A. Zink, L. Codecá, and S. Clarke, "A digital twin smart city for citizen feedback," *Cities*, vol. 110, Mar. 2021, Art. no. 103064.

- [71] A. Gegov, "Complex systems modelling by rule based networks," in *Proc. 8th Wseas Int. Conf. Artif. Intell., Knowl. Eng. Data Bases*, 2009, p. 122.
- [72] W. Waites, M. Cavaliere, D. Manheim, J. Panovska-Griffiths, and V. Danos, "Rule-based epidemic models," *J. Theor. Biol.*, vol. 530, Dec. 2021, Art. no. 110851.
- [73] H. Liu, A. Gegov, and F. Stahl, "Categorization and construction of rule based systems," in *Engineering Applications of Neural Networks (EANN)*, vol. 459. Berlin, Germany: Springer-Verlag, 2014, pp. 183–194.
- [74] V. Dimitrova, M. O. Mehmood, D. Thakker, B. Sage-Vallier, J. Valdes, and A. G. Cohn, "An ontological approach for pathology assessment and diagnosis of tunnels," *Eng. Appl. Artif. Intell.*, vol. 90, Apr. 2020, Art. no. 103450.
- [75] G. Yu, Y. Wang, Z. Mao, M. Hu, V. Sugumaran, and Y. K. Wang, "A digital twin-based decision analysis framework for operation and maintenance of tunnels," *Tunnelling Underground Space Technol.*, vol. 116, Oct. 2021, Art. no. 104125.
- [76] W. Z. Taffese and E. Sistonon, "Machine learning for durability and service-life assessment of reinforced concrete structures: Recent advances and future directions," *Automat. Construct.*, vol. 77, pp. 1–14, May 2017.
- [77] Z. Pang, J. Guo, W. Sun, Y. Xiao, and M. Yu, "Cross-domain person re-identification by hybrid supervised and unsupervised learning," *Int. J. Speech Technol.*, vol. 52, no. 3, pp. 2987–3001, Feb. 2022.
- [78] R. Reddy and G. K. Shyam, "Analysis through machine learning techniques: A survey," in *Proc. 2nd Int. Conf. Green Comput. Internet Things (ICGCIoT)*, Aug. 2018, pp. 542–546.
- [79] P. P. Shinde and S. Shah, "A review of machine learning and deep learning applications," in *Proc. 4th Int. Conf. Comput. Commun. Control Autom. (ICCUBE)*, Aug. 2018, pp. 1–6.
- [80] J. Ren, B. Zhang, X. Zhu, and S. Li, "Damaged cable identification in cable-stayed bridge from bridge deck strain measurements using support vector machine," *Adv. Struct. Eng.*, vol. 25, no. 4, pp. 754–771, Mar. 2022.
- [81] H. Naraniecki, R. Hartung, S. Marx, and K. Klemt-Albert, "Zustandsprognose von ingenieurbauwerken auf basis von digitalen zwillingen und bestandsdaten," *Bautechnik*, vol. 99, no. 3, pp. 173–181, Mar. 2022.
- [82] Y. N. Imamverdiyev and F. J. Abdullayeva, "Deep learning in cyber-security: Challenges and approaches," *Int. J. Cyber Warfare Terrorism*, vol. 10, no. 2, pp. 82–105, Apr./Jun. 2020.
- [83] J. Wang, Y. Ma, L. Zhang, R. X. Gao, and D. Wu, "Deep learning for smart manufacturing: Methods and applications," *J. Manuf. Syst.*, vol. 48, pp. 144–156, Jul. 2018.
- [84] W. Fan et al., "Machine learning applied to the design and inspection of reinforced concrete bridges: Resilient methods and emerging applications," *Structures*, vol. 33, pp. 3954–3963, Oct. 2021.
- [85] D. Feng and M. Q. Feng, "Computer vision for SHM of civil infrastructure: From dynamic response measurement to damage detection—A review," *Eng. Struct.*, vol. 156, pp. 105–117, Feb. 2018.
- [86] Y. Xiang, B. Pan, and L. Luo, "A new model updating strategy with physics-based and data-driven models," *Struct. Multidisciplinary Optim.*, vol. 64, no. 1, pp. 163–176, Jul. 2021.
- [87] H. Kim, S. Cho, and S.-H. Sim, "Data fusion of acceleration and angular velocity for improved model updating," *Measurement*, vol. 91, pp. 239–250, Sep. 2016.
- [88] C. Rao, H. Sun, and Y. Liu, "Physics-informed deep learning for incompressible laminar flows," *Theor. Appl. Mech. Lett.*, vol. 10, no. 3, pp. 207–212, 2020.
- [89] Z. Chen, R. Zhang, J. Zheng, and H. Sun, "Sparse Bayesian learning for structural damage identification," *Mech. Syst. Signal Process.*, vol. 140, Jun. 2020, Art. no. 106689.
- [90] Y. Yang et al., "Estimation of full-field, full-order experimental modal model of cable vibration from digital video measurements with physics-guided unsupervised machine learning and computer vision," *Struct. Control Health Monit.*, vol. 26, no. 6, p. e2358, 2019.
- [91] X. Zhou, J. Wang, M. Guo, and Z. Gao, "Cross-platform online visualization system for open BIM based on WebGL," *Multimedia Tools Appl.*, vol. 78, no. 20, pp. 28575–28590, Oct. 2019.
- [92] L. Nie, G. Hou, W. Li, and P. Li, "Research on automatic visualization technology of bridge disease information based on BIM," in *Proc. Int. Conf. Intell. Traffic Syst. Smart City (ITSSC)*. Bellingham, WA, USA: SPIE, vol. 12165, 2022, Art. no. 1216505.
- [93] X.-S. Chen, C.-C. Liu, and I.-C. Wu, "A BIM-based visualization and warning system for fire rescue," *Adv. Eng. Informat.*, vol. 37, pp. 42–53, Aug. 2018.
- [94] Z. Liu, X. Gu, Q. Dong, S. Tu, and S. Li, "3D visualization of airport pavement quality based on BIM and WebGL integration," *J. Transp. Eng., B, Pavements*, vol. 147, no. 3, 2021, Art. no. 04021024.
- [95] B. Keskin, B. Salman, and O. Koseoglu, "Architecting a BIM-based digital twin platform for airport asset management: A model-based system engineering with SysML approach," *J. Construct. Eng. Manage.*, vol. 148, no. 5, May 2022.
- [96] Z. X. Zhu, C. Liu, and X. Xu, "Visualisation of the digital twin data in manufacturing by using augmented reality," *Proc. CIRP*, vol. 81, pp. 898–903, Jan. 2019.
- [97] P. Le Noury, R. Polman, M. Maloney, and A. Gorman, "A narrative review of the current state of extended reality technology and how it can be utilised in sport," *Sports Med.*, vol. 52, no. 7, pp. 1473–1489, Jul. 2022.
- [98] Y. Ham and J. Kim, "Participatory sensing and digital twin city: Updating virtual city models for enhanced risk-informed decision-making," *J. Manag. Eng.*, vol. 36, no. 3, 2020, Art. no. 04020005.
- [99] Y. K. Liu, S. K. Ong, and A. Y. C. Nee, "State-of-the-art survey on digital twin implementations," *Adv. Manuf.*, vol. 10, no. 1, pp. 1–23, 2022.
- [100] U. Riedlinger et al., "Mixed reality support for bridge inspectors," in *Proc. IEEE Conf. Virtual Reality 3D User Interfaces Abstr. Workshops (VRW)*, Jun. 2022, pp. 695–696.
- [101] F. Tanaka, M. Tsuchida, and M. Onosato, "Associating 2D sketch information with 3D CAD models for VR/AR viewing during bridge maintenance process," *Int. J. Autom. Technol.*, vol. 13, no. 4, pp. 482–489, Jul. 2019.
- [102] R. Hajdin, "Managing existing bridges—On the brink of an exciting future," in *Maintenance, Safety, Risk, Management and Life-Cycle Performance of Bridges*. Boca Raton, FL, USA: CRC Press, 2018, pp. 70–87.
- [103] F. Jiang, Y. Ding, Y. Song, F. Geng, and Z. Wang, "Digital twin-driven framework for fatigue life prediction of steel bridges using a probabilistic multiscale model: Application to segmental orthotropic steel deck specimen," *Eng. Struct.*, vol. 241, Aug. 2021, Art. no. 112461.
- [104] S. Kaewunruen, J. Sresakoolchai, W. Ma, and O. Phil-Ebosie, "Digital twin aided vulnerability assessment and risk-based maintenance planning of bridge infrastructures exposed to extreme conditions," *Sustainability*, vol. 13, no. 4, p. 2051, Feb. 2021.
- [105] J.-S. Kang, K. Chung, and E. J. Hong, "Multimedia knowledge-based bridge health monitoring using digital twin," *Multimedia Tools Appl.*, vol. 80, nos. 26–27, pp. 34609–34624, Nov. 2021.
- [106] M. S. Mafipour, S. Vilgertshofer, and A. Borrmann, "Automated geometric digital twinning of bridges from segmented point clouds by parametric prototype models," *Autom. Construct.*, vol. 156, Dec. 2023, Art. no. 105101.
- [107] K. Lin, Y.-L. Xu, X. Lu, Z. Guan, and J. Li, "Digital twin-based collapse fragility assessment of a long-span cable-stayed bridge under strong earthquakes," *Autom. Construct.*, vol. 123, Mar. 2021, Art. no. 103547.
- [108] F. Ghahari, N. Malekghani, H. Ebrahimian, and E. Taciroglu, "Bridge digital twinning using an output-only Bayesian model updating method and recorded seismic measurements," *Sensors*, vol. 22, no. 3, p. 1278, Feb. 2022.
- [109] A. Shabani and M. Kioumars, "Seismic assessment and strengthening of a historical masonry bridge considering soil-structure interaction," *Eng. Struct.*, vol. 293, Oct. 2023, Art. no. 116589.
- [110] K. Lin, Y.-L. Xu, X. Lu, Z. Guan, and J. Li, "Digital twin-based life-cycle seismic performance assessment of a long-span cable-stayed bridge," *Bull. Earthq. Eng.*, vol. 21, no. 2, pp. 1203–1227, Jan. 2023.
- [111] M. Hu, Y. Liu, V. Sugumaran, B. Liu, and J. Du, "Automated structural defects diagnosis in underground transportation tunnels using semantic technologies," *Autom. Construct.*, vol. 107, Nov. 2019, Art. no. 102929.
- [112] Z. Wu, Y. Chang, Q. Li, and R. Cai, "A novel method for tunnel digital twin construction and virtual-real fusion application," *Electronics*, vol. 11, no. 9, p. 1413, Apr. 2022.
- [113] R. Sanfilippo et al., "The digitalization of the tunnel Schöneich-using a digital twin to improve operation and maintenance workflows of an existing road tunnel," in *Expanding Underground-Knowledge and Passion to Make a Positive Impact on the World*. Boca Raton, FL, USA: CRC Press, 2023, pp. 2877–2884.
- [114] J. Vieira, J. Poças Martins, N. Marques de Almeida, H. Patrício, and J. Gomes Morgado, "Towards resilient and sustainable rail and road networks: A systematic literature review on digital twins," *Sustainability*, vol. 14, no. 12, p. 7060, Jun. 2022.

- [115] W. J. Steyn and A. Broekman, "Development of a digital twin of a local road network: A case study," *J. Test. Eval.*, vol. 50, no. 6, pp. 2901–2915, Nov. 2022.
- [116] F. D'Amico, L. Bianchini Ciampoli, A. Di Benedetto, L. Bertolini, and A. Napolitano, "Integrating non-destructive surveys into a preliminary BIM-oriented digital model for possible future application in road pavements management," *Infrastructures*, vol. 7, no. 1, p. 10, Jan. 2022.
- [117] T. Cao, Y. Wang, and S. Liu, "Pavement crack detection based on 3D edge representation and data communication with digital twins," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 7, pp. 7697–7706, Aug. 2023.
- [118] L. Barisic, E. Levenberg, A. Skar, A. Boyd, and P. Zoulis, "A thermal digital twin for condition monitoring of asphalt roads," in *Green and Intelligent Technologies for Sustainable and Smart Asphalt Pavements*. Boca Raton, FL, USA: CRC Press, 2021, pp. 709–713.
- [119] G. Sotiriadis, T. Kotzakolios, V. Kostopoulos, and M. Gemou, "A digital twin assisted and embedded strain gauge monitoring system," *Transp. Res. Proc.*, vol. 72, pp. 2038–2045, Jan. 2023.
- [120] A. Kampczyk and K. Dybel, "The fundamental approach of the digital twin application in railway turnouts with innovative monitoring of weather conditions," *Sensors*, vol. 21, no. 17, p. 5757, Aug. 2021.
- [121] C. Sierra, S. Paul, A. Rahman, and A. Kulkarni, "Development of a cognitive digital twin for pavement infrastructure health monitoring," *Infrastructures*, vol. 7, no. 9, p. 113, Aug. 2022.
- [122] Y. Zhengning and Z. Yongli, "An evolution-simulation-verification framework of aviation products in the future," in *Proc. IEEE 2nd Int. Conf. Civil Aviation Saf. Inf. Technol. (ICCASIT)*, Oct. 2020, pp. 1004–1007.
- [123] Z. Gui et al., "A cloud-edge-terminal-based robotic system for airport runway inspection," *Ind. Robot: Int. J. Robot. Res. Appl.*, vol. 48, no. 6, pp. 846–855, Nov. 2021.
- [124] Q. Shao, M. Yang, C. Xu, H. Wang, and H. Liu, "Fire risk analysis of runway excursion accidents in high-plateau airport," *IEEE Access*, vol. 8, pp. 204400–204416, 2020.
- [125] Z. Li, P. Wang, X. Han, and D. Shang, "Research on digital twin technology of port structure safety based on floating 3D display," in *Proc. IEEE 3rd Int. Conf. Inf. Technol., Big Data Artif. Intell. (ICIBA)*, vol. 3, May 2023, pp. 976–981.
- [126] Z. Wu, C. Ren, X. Wu, L. Wang, L. Zhu, and Z. Lv, "Research on digital twin construction and safety management application of inland waterway based on 3D video fusion," *IEEE Access*, vol. 9, pp. 109144–109156, 2021.
- [127] Y. Yigit et al., "TwinPort: 5G drone-assisted data collection with digital twin for smart seaports," *Sci. Rep.*, vol. 13, no. 1, p. 12310, 2023.



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